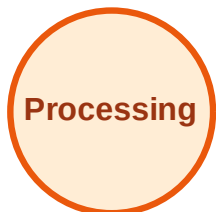


# BFS Traversal

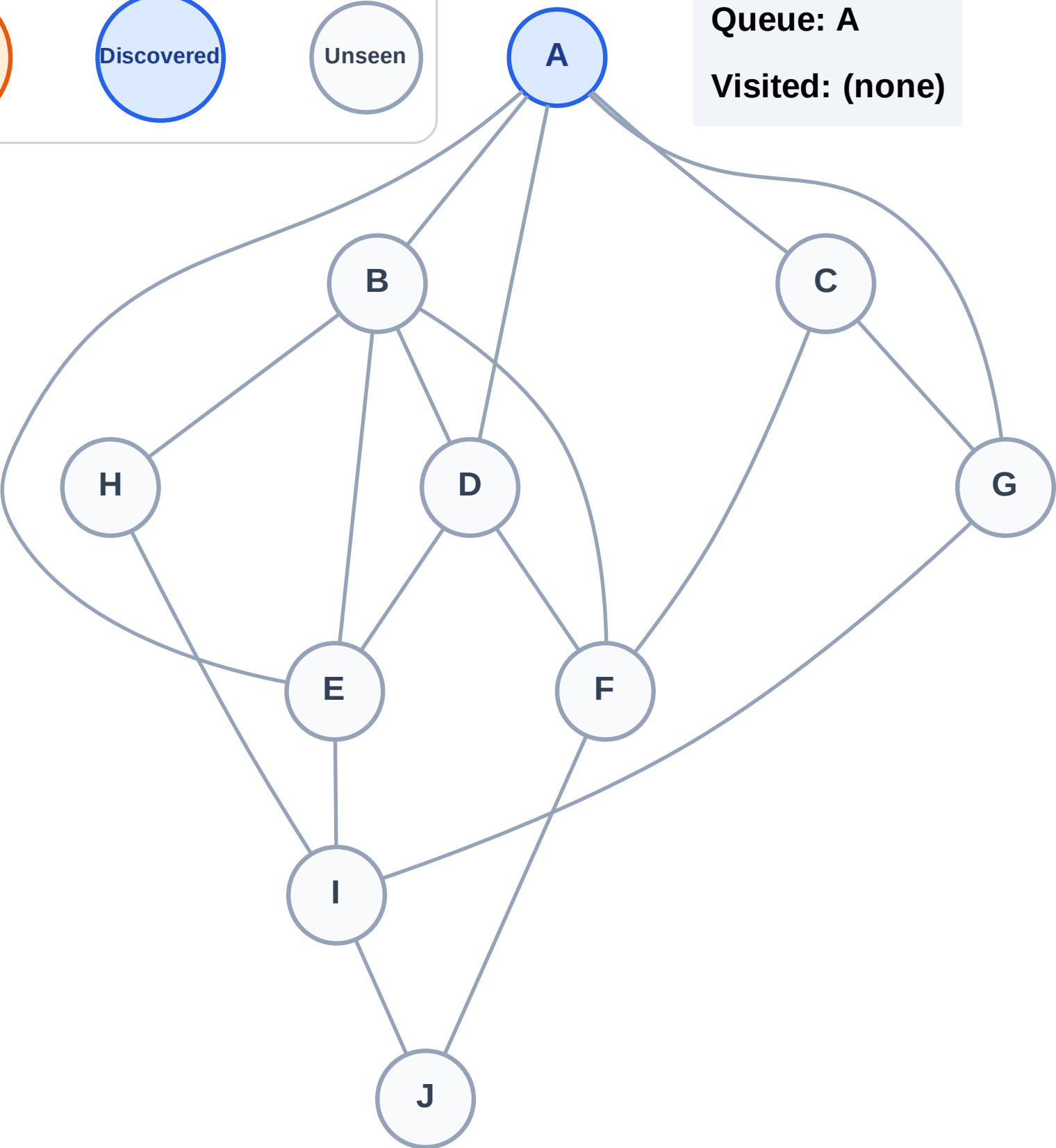
Step 1: Start at A

## Legend



Queue: A

Visited: (none)



# BFS Traversal

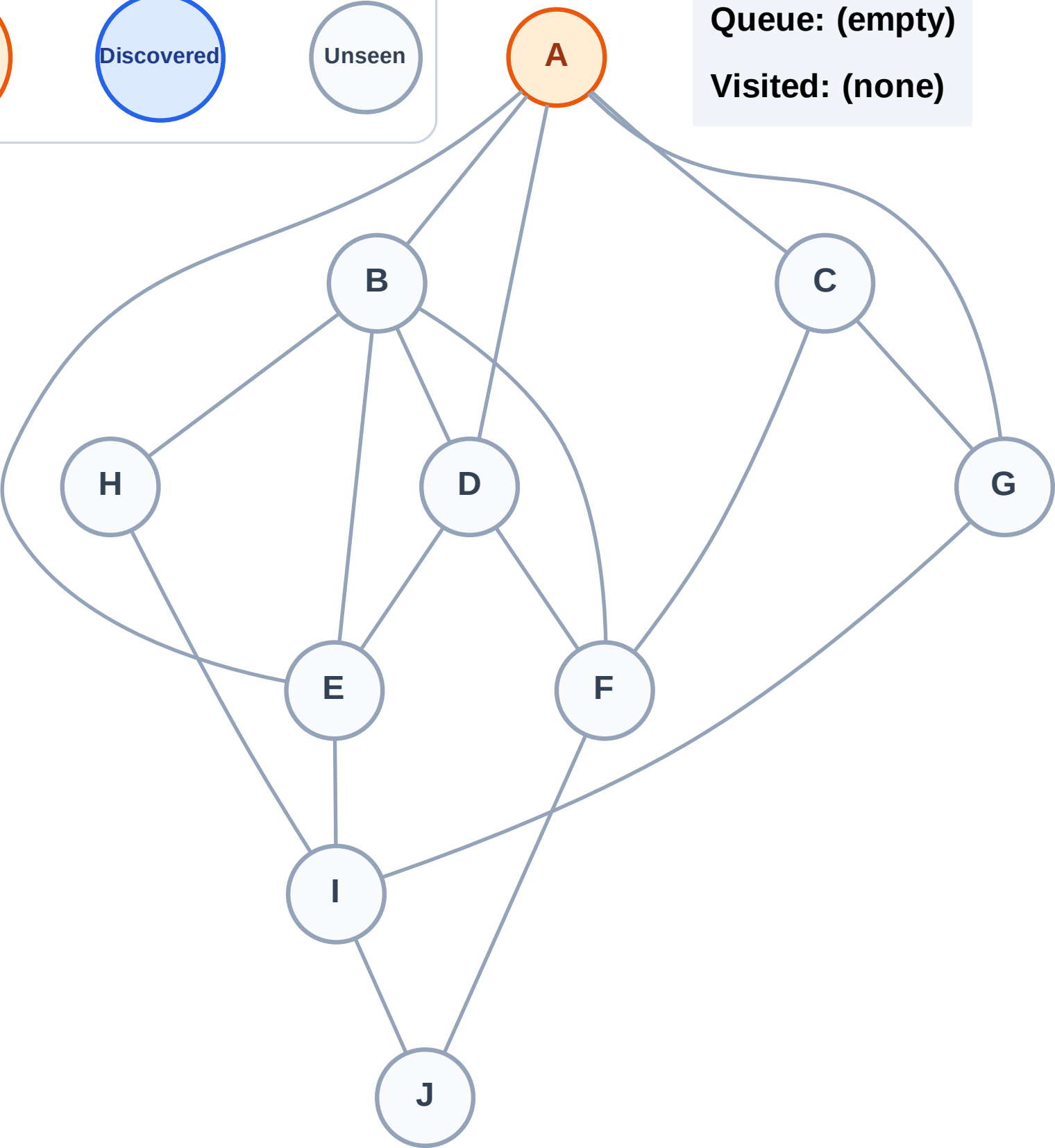
Step 2: Process node A

## Legend



Queue: (empty)

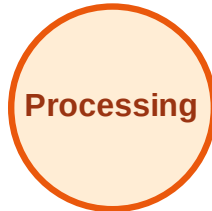
Visited: (none)



# BFS Traversal

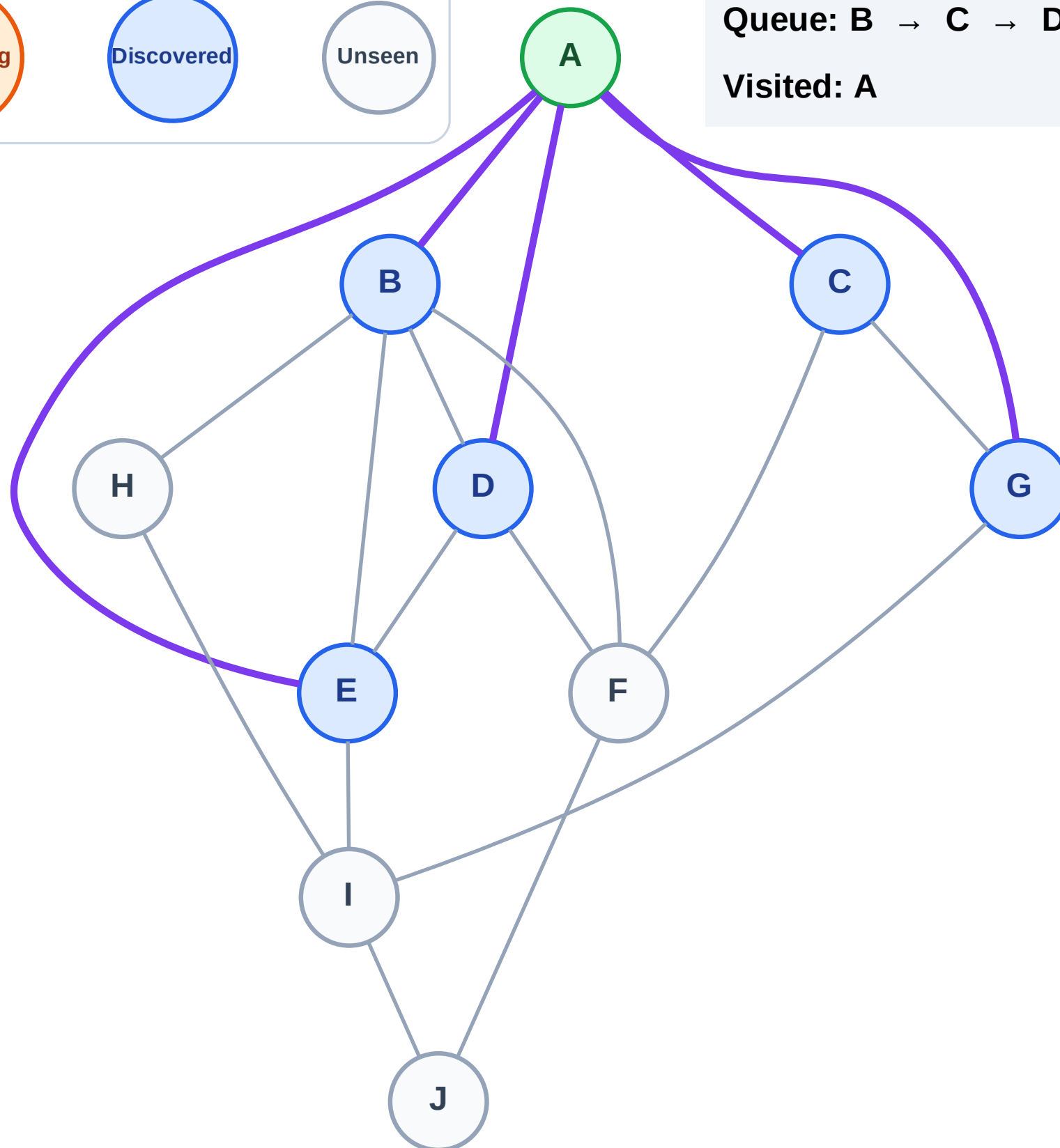
Step 3: Discover B, C, D, E, G from A

## Legend



Queue: B → C → D → E → G

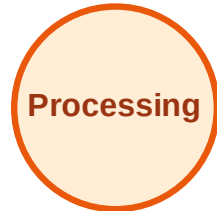
Visited: A



# BFS Traversal

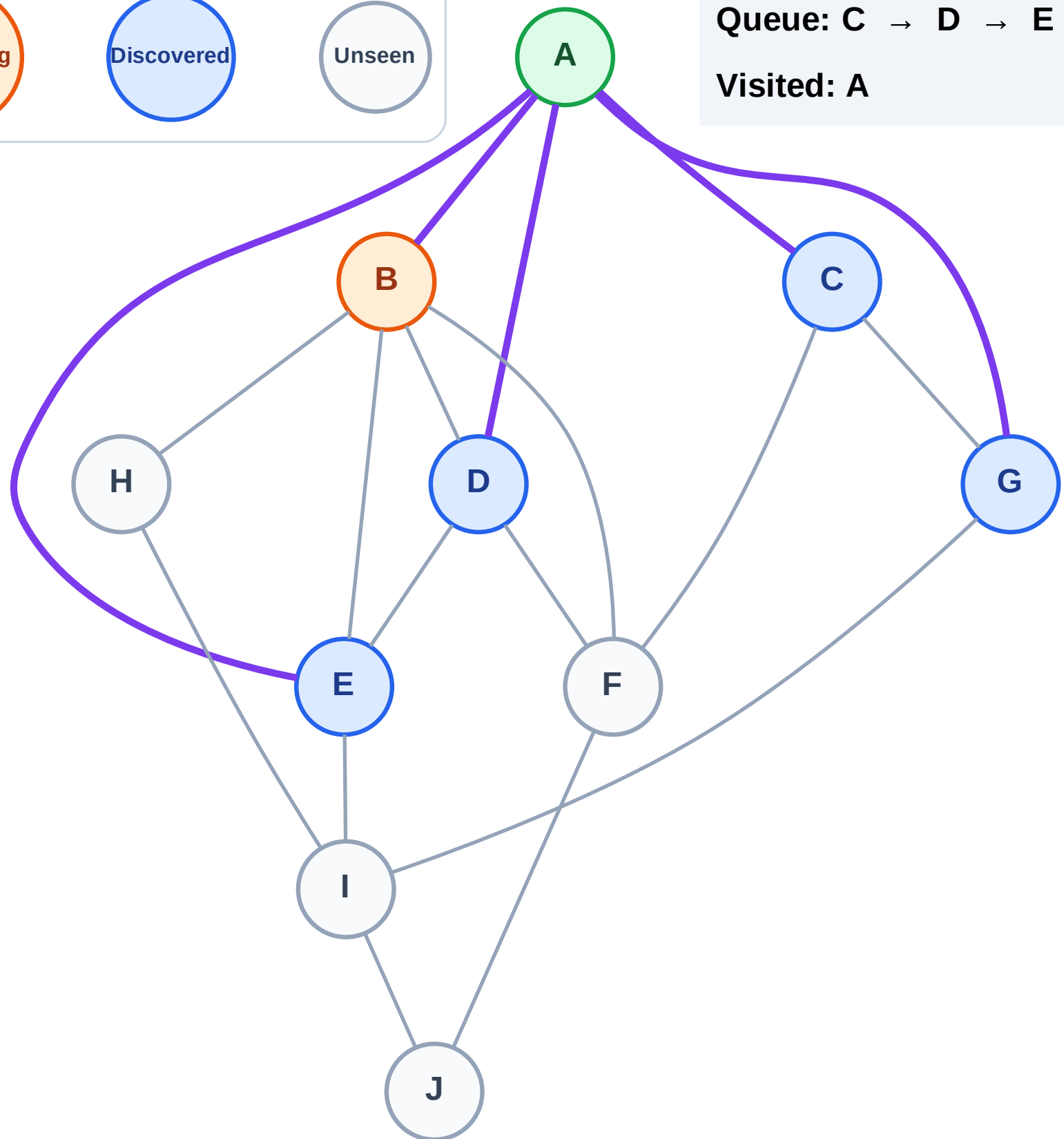
Step 4: Process node B

## Legend



Queue: C → D → E → G

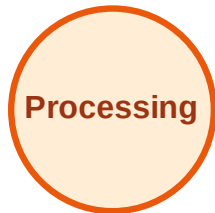
Visited: A



# BFS Traversal

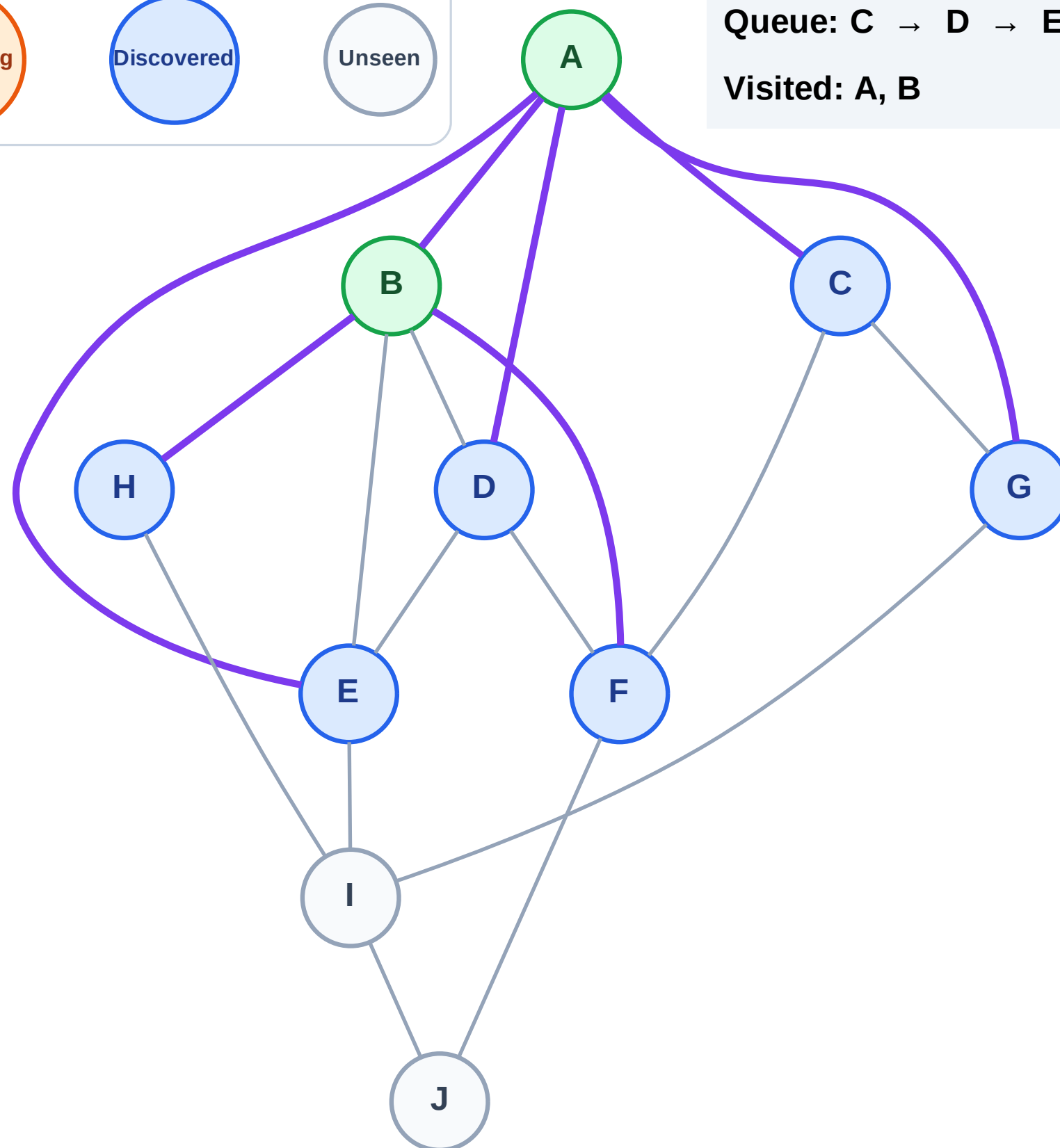
Step 5: Discover F, H from B

## Legend



Queue: C → D → E → G → F → H

Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

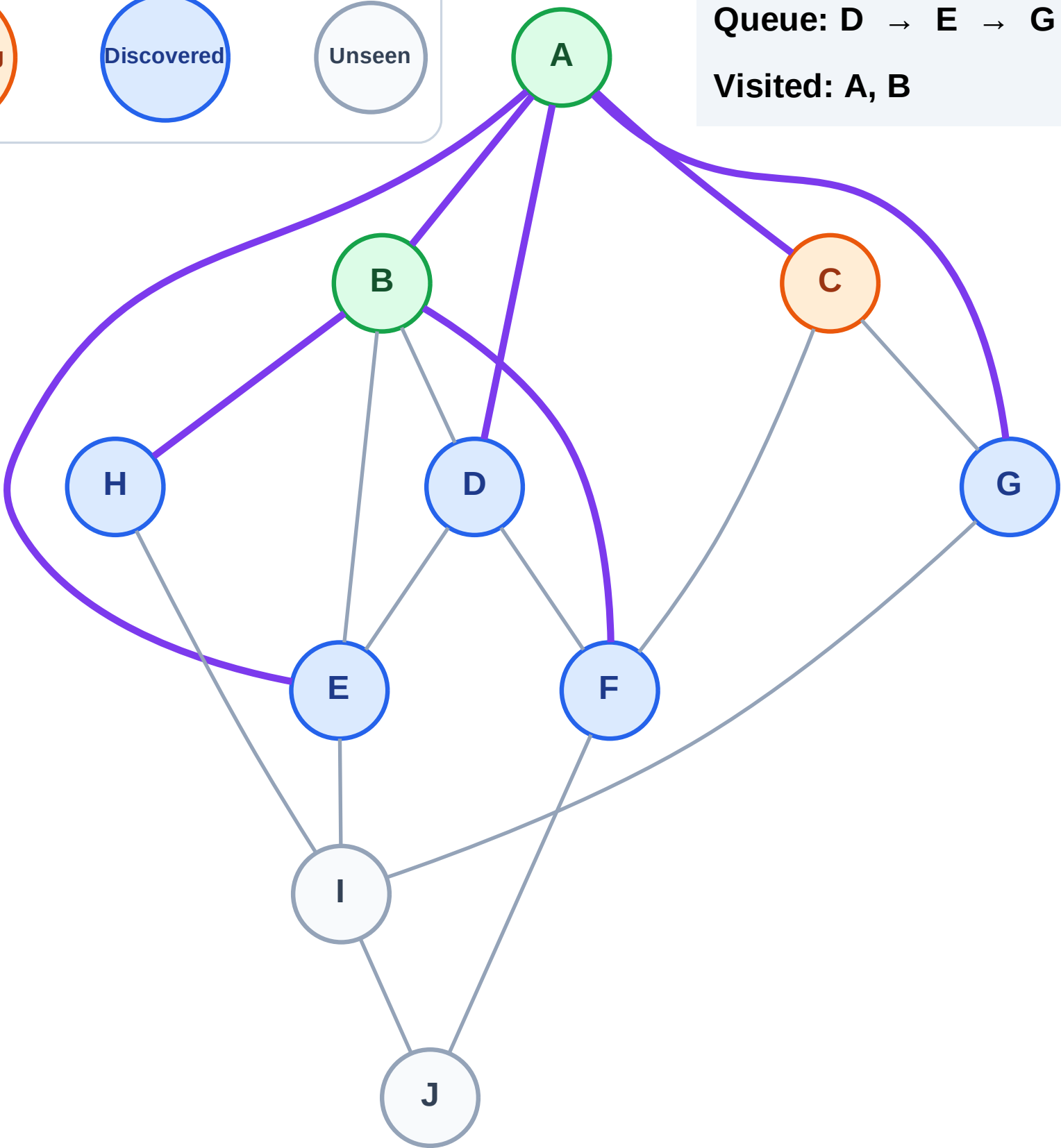
Processing

Discovered

Unseen

Queue: D → E → G → F → H

Visited: A, B



# BFS Traversal

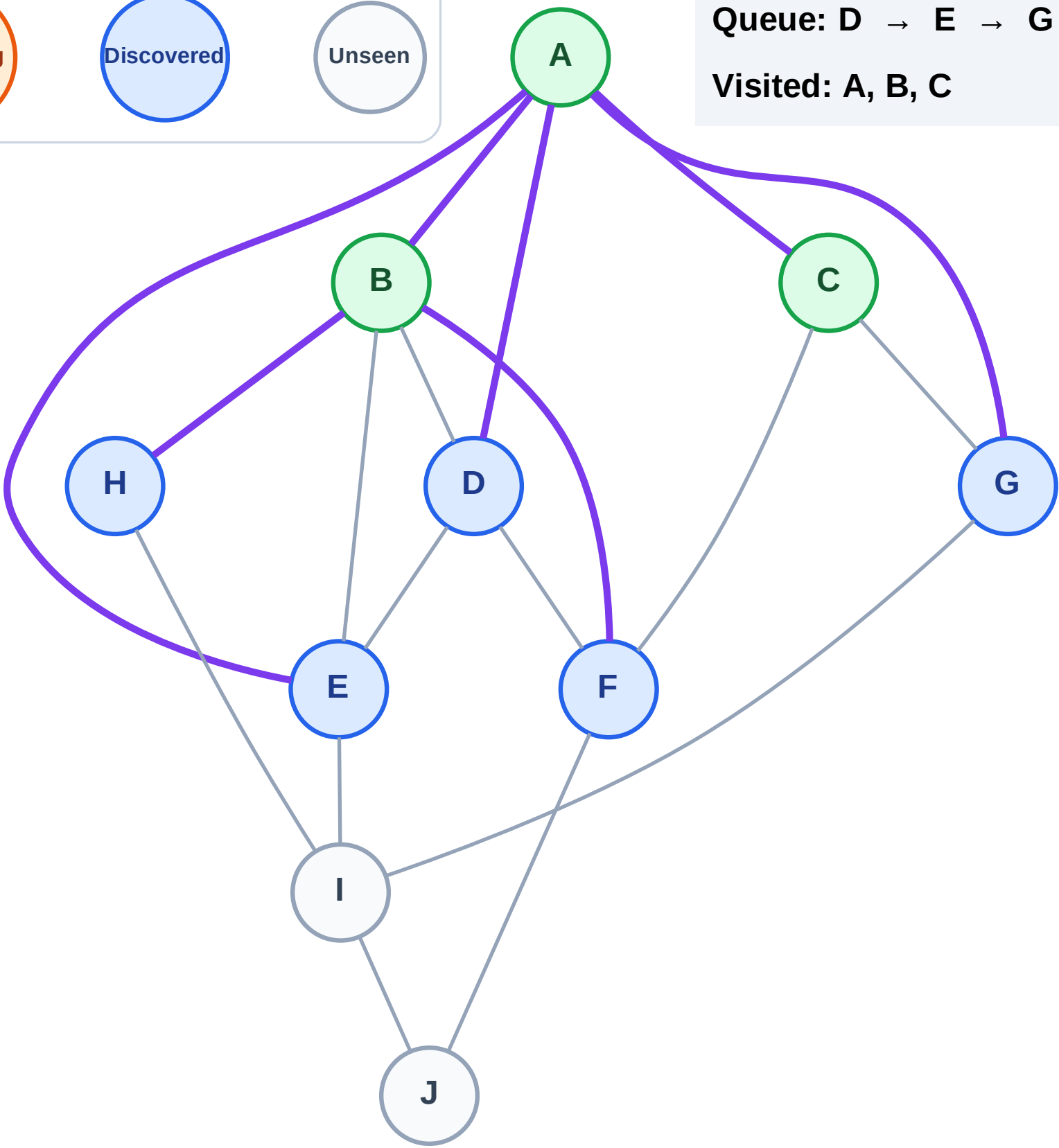
Step 7: No new neighbors from C

## Legend



Queue: D → E → G → F → H

Visited: A, B, C

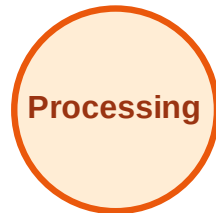




# BFS Traversal

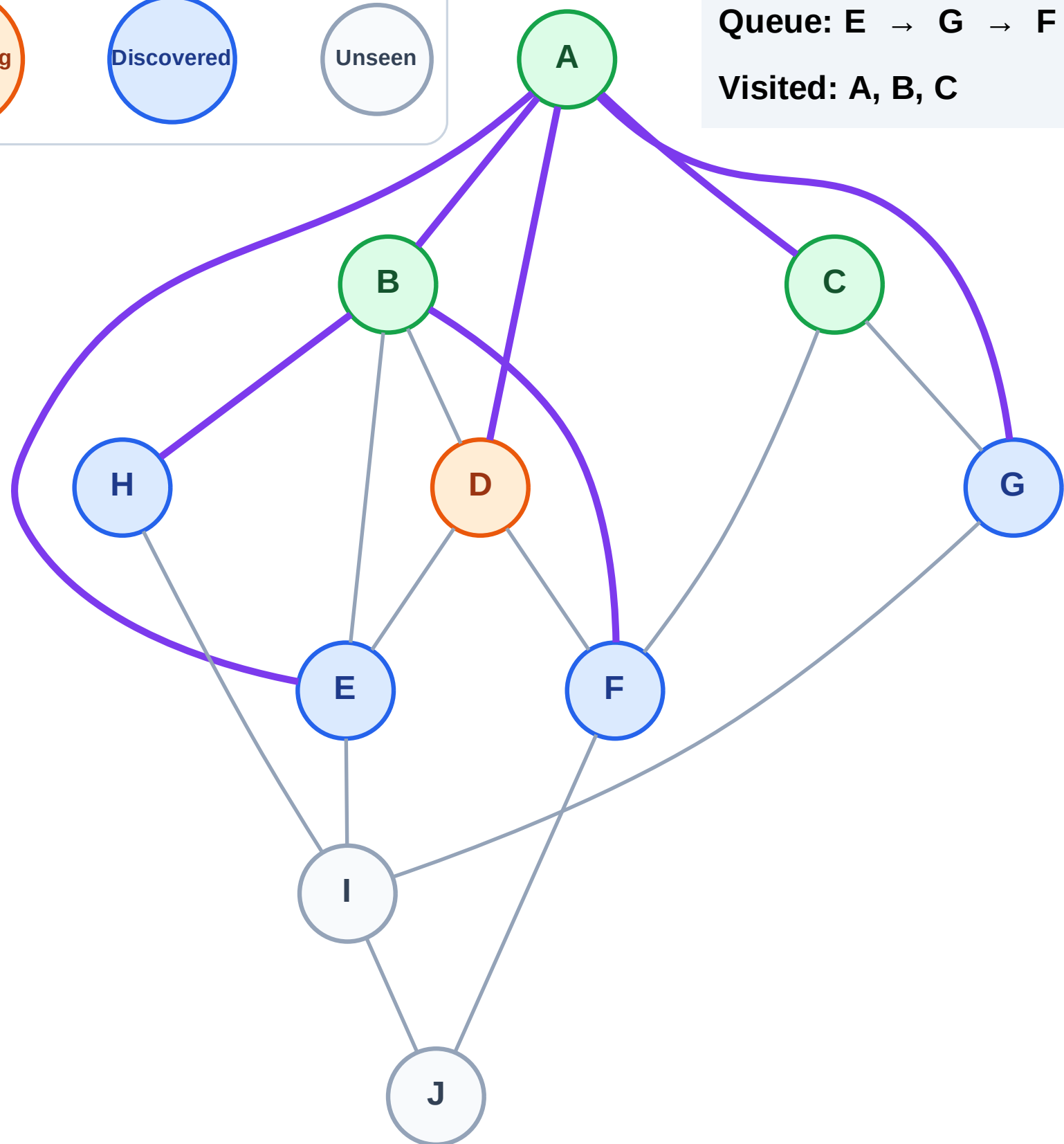
Step 8: Process node D

## Legend



Queue: E → G → F → H

Visited: A, B, C

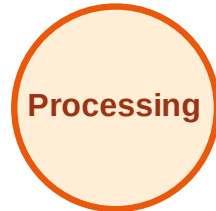




# BFS Traversal

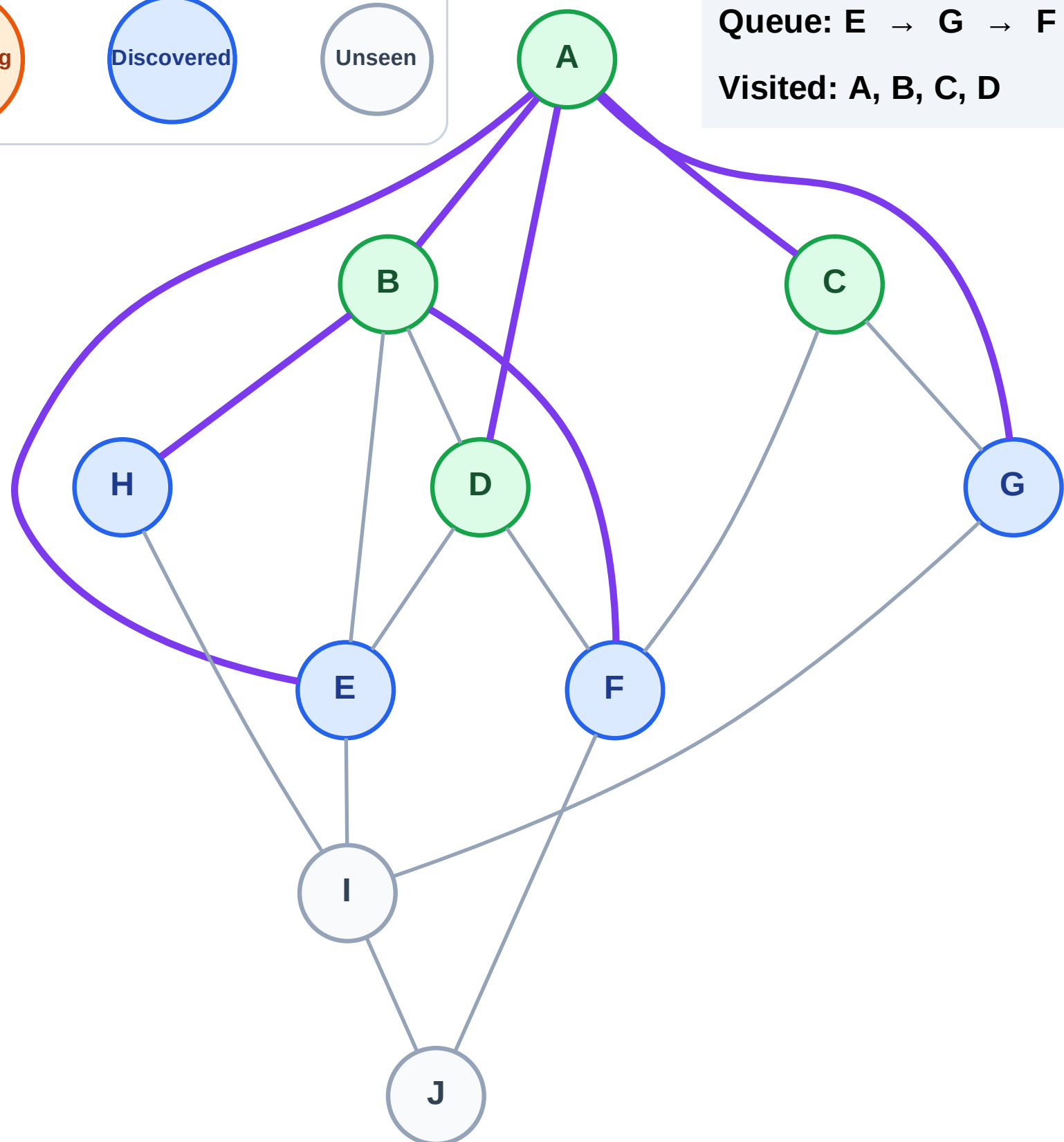
Step 9: No new neighbors from D

## Legend



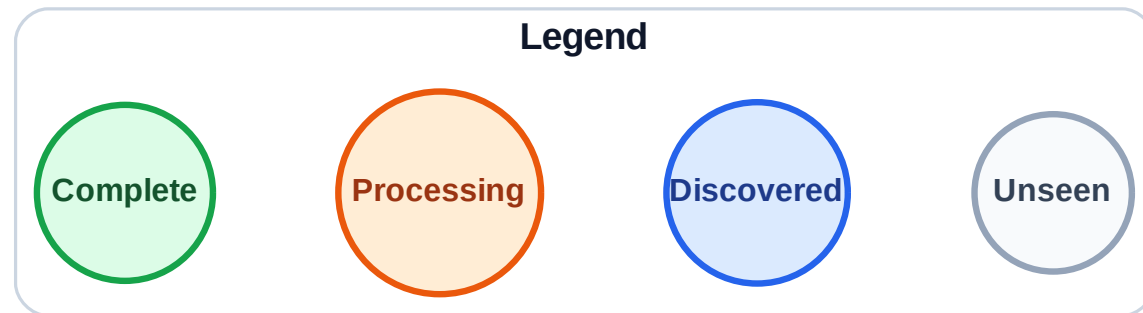
Queue: E → G → F → H

Visited: A, B, C, D

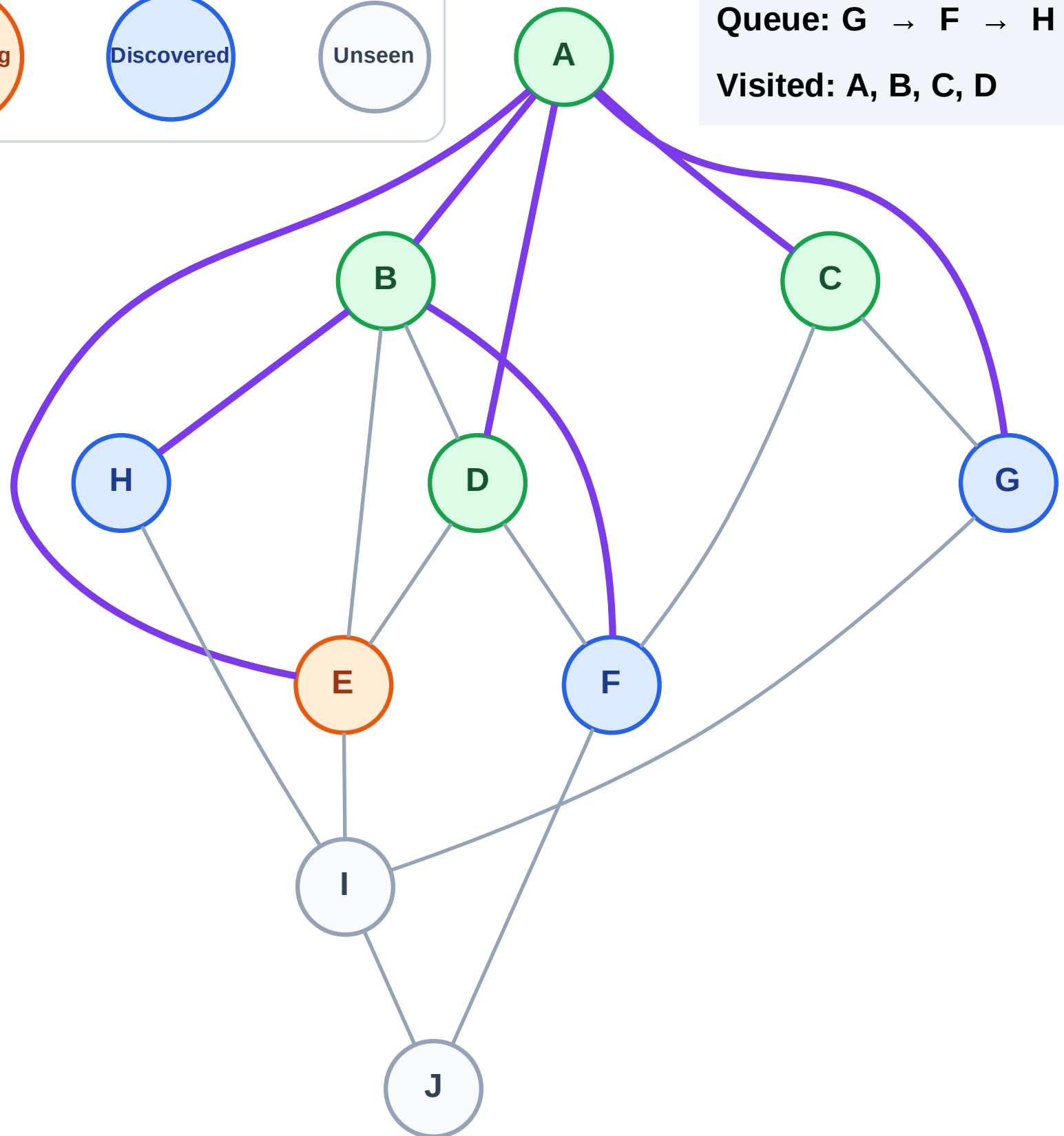


# BFS Traversal

Step 10: Process node E



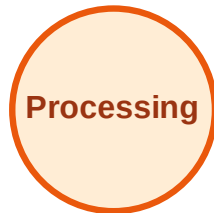
Queue: G → F → H  
Visited: A, B, C, D



# BFS Traversal

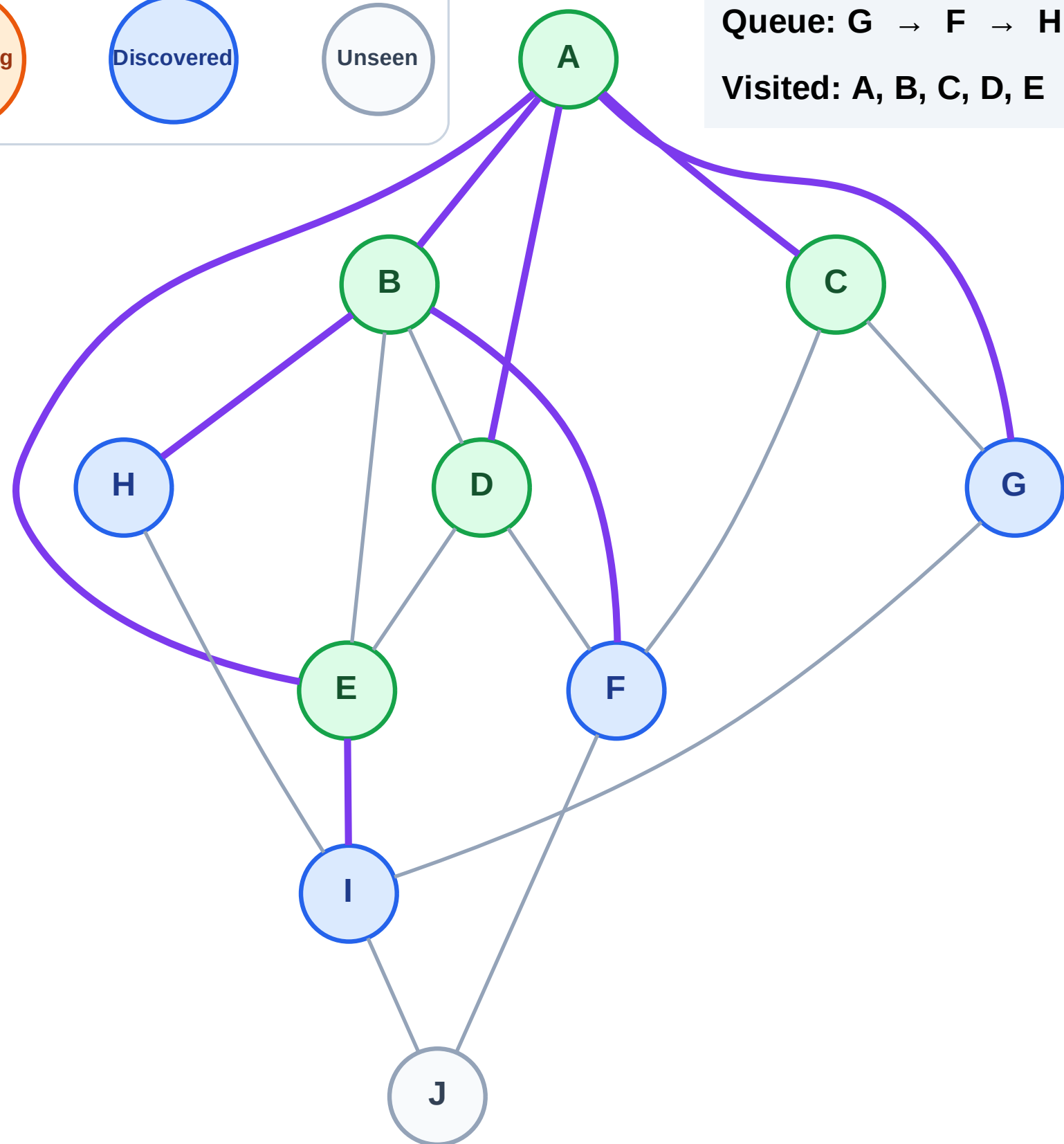
Step 11: Discover I from E

## Legend



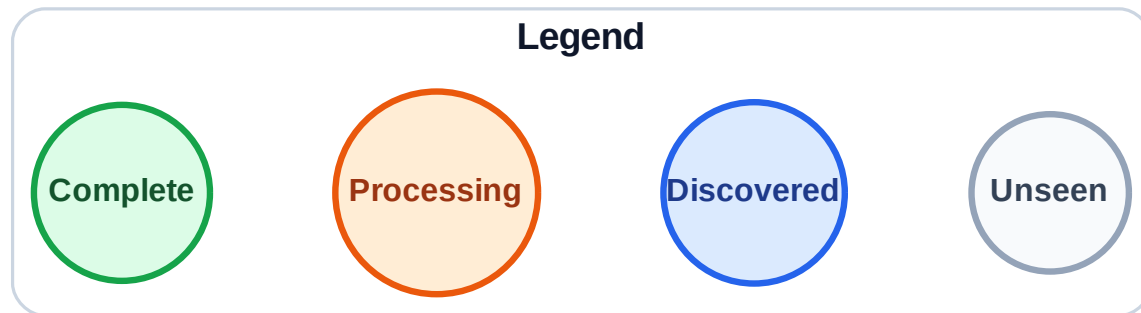
Queue: G → F → H → I

Visited: A, B, C, D, E

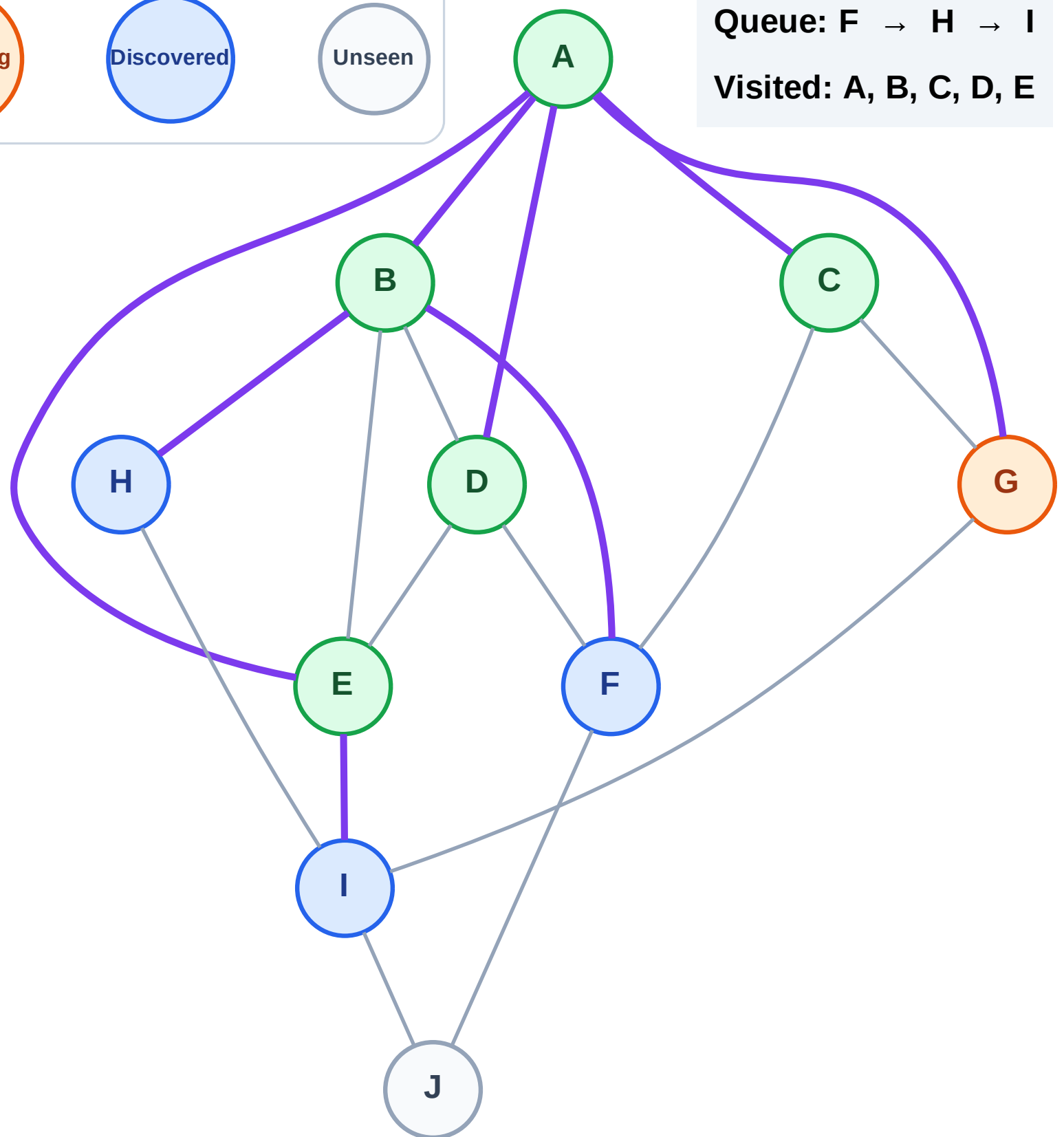


# BFS Traversal

Step 12: Process node G



Queue: F → H → I  
Visited: A, B, C, D, E



# BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

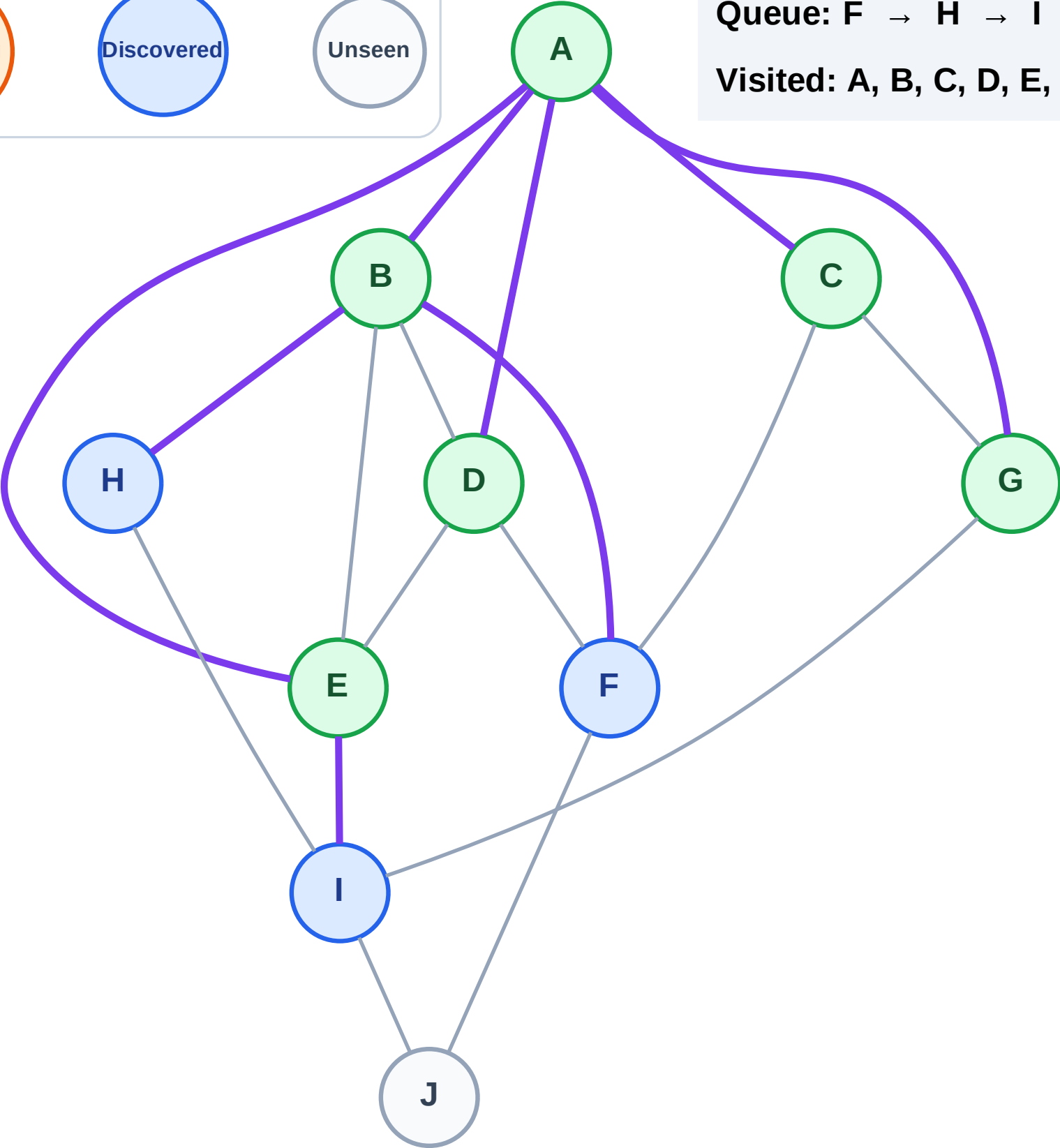
Processing

Discovered

Unseen

Queue: F → H → I

Visited: A, B, C, D, E, G



# BFS Traversal

Step 14: Process node F

Legend

Complete

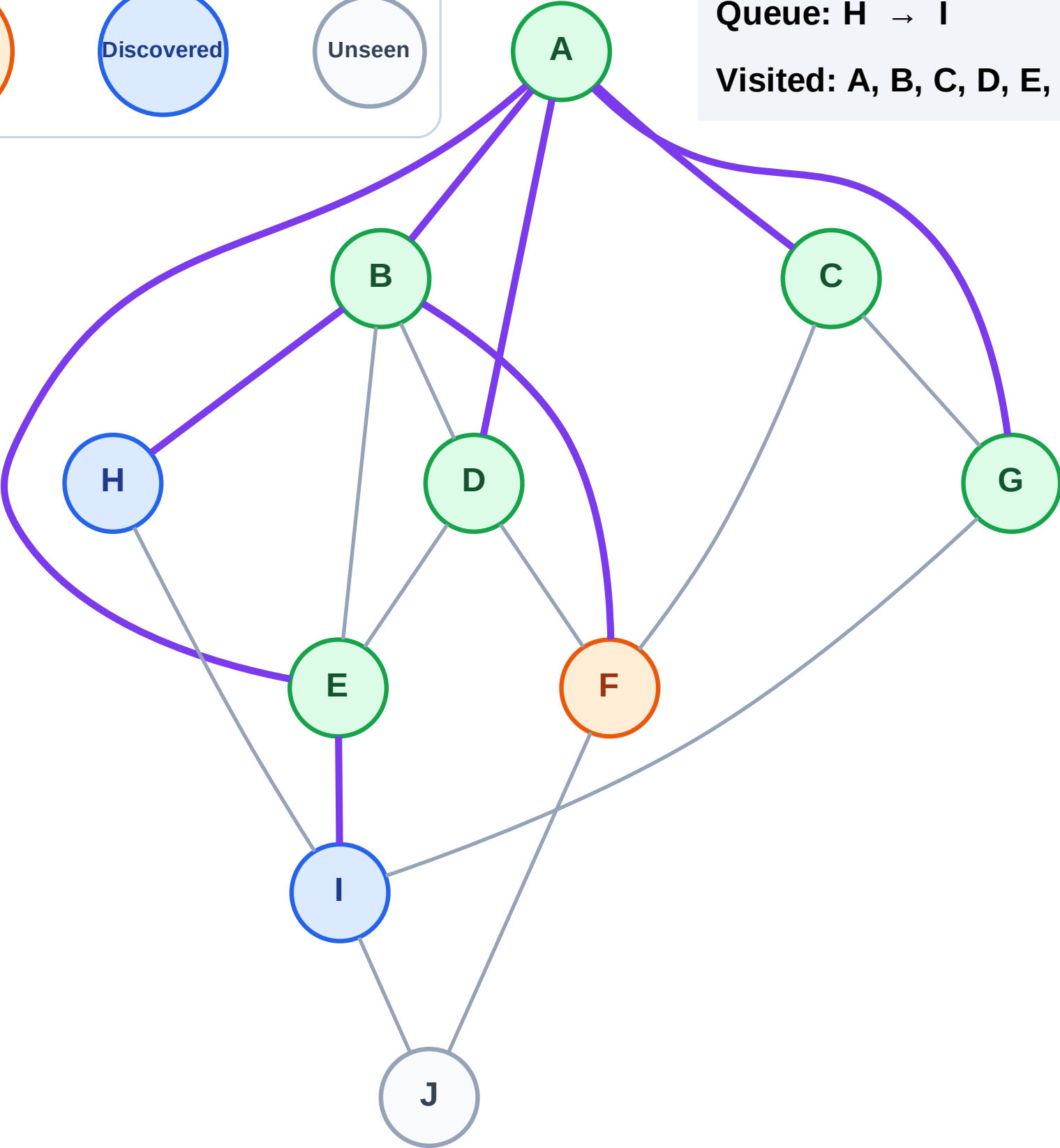
Processing

Discovered

Unseen

Queue: H → I

Visited: A, B, C, D, E, G



# BFS Traversal

Step 15: Discover J from F

Legend

Complete

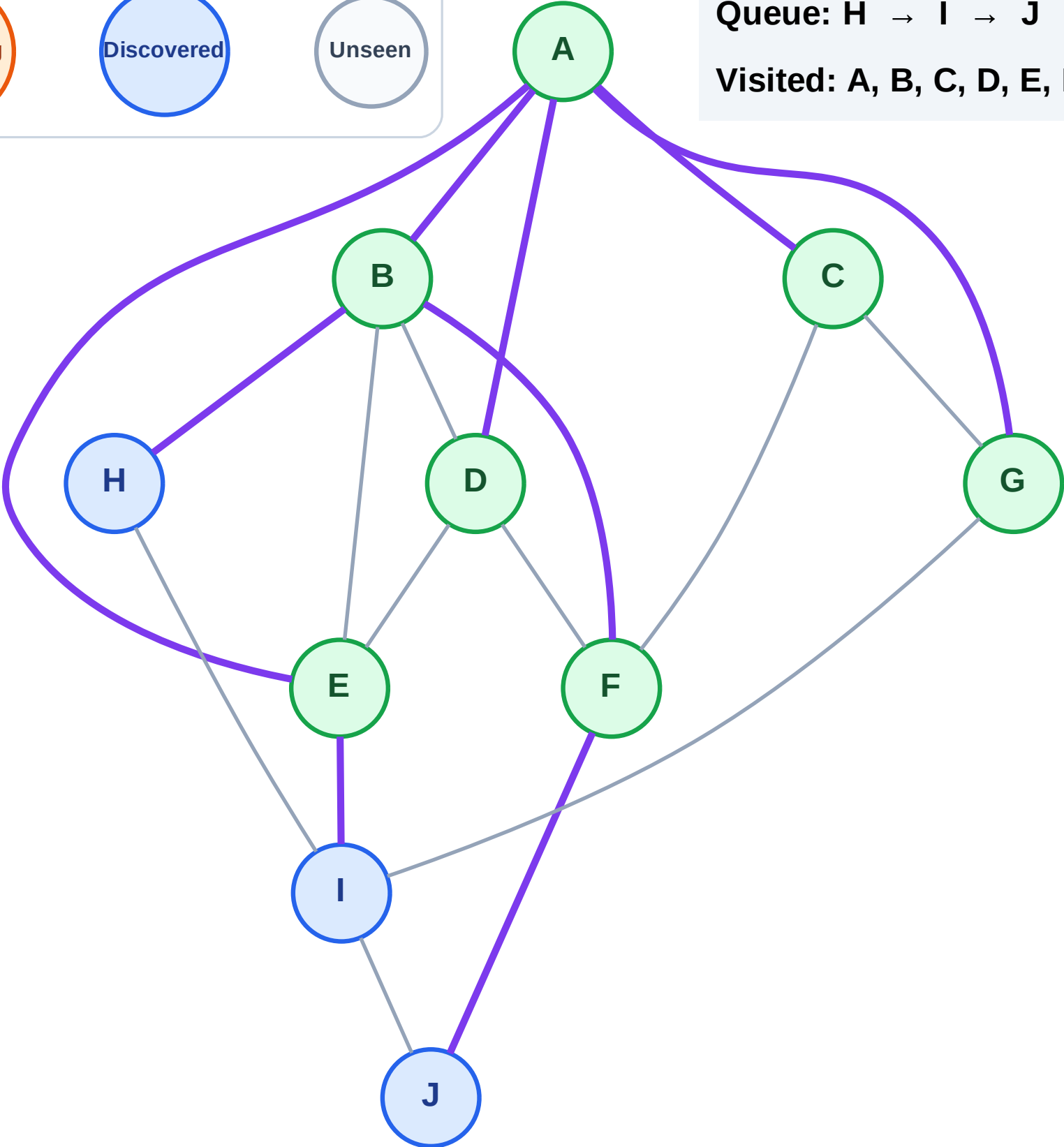
Processing

Discovered

Unseen

Queue: H → I → J

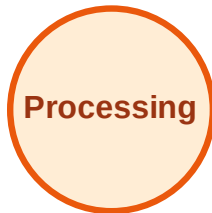
Visited: A, B, C, D, E, F, G



# BFS Traversal

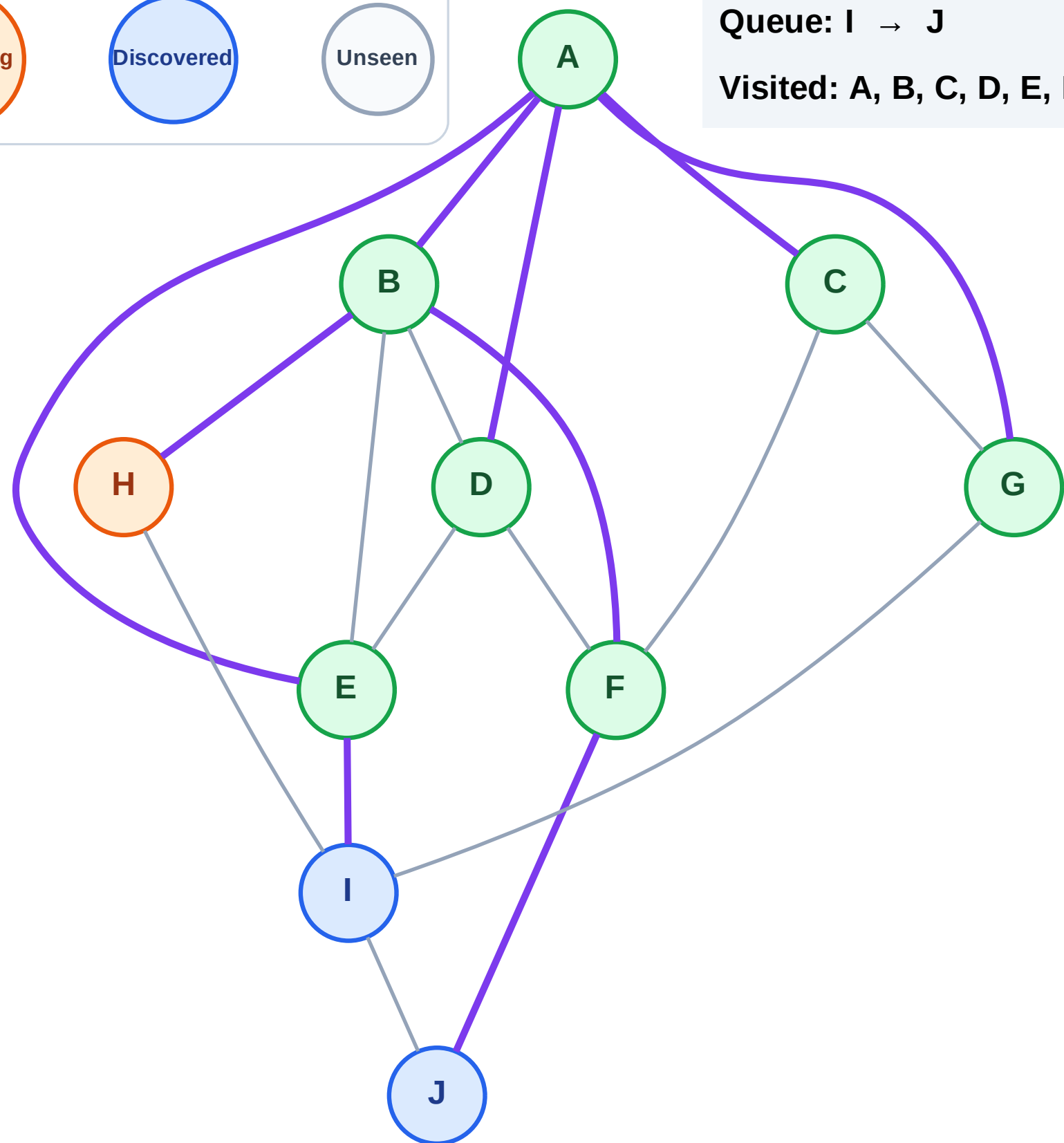
Step 16: Process node H

## Legend



Queue: I → J

Visited: A, B, C, D, E, F, G

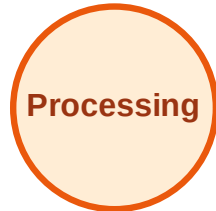




# BFS Traversal

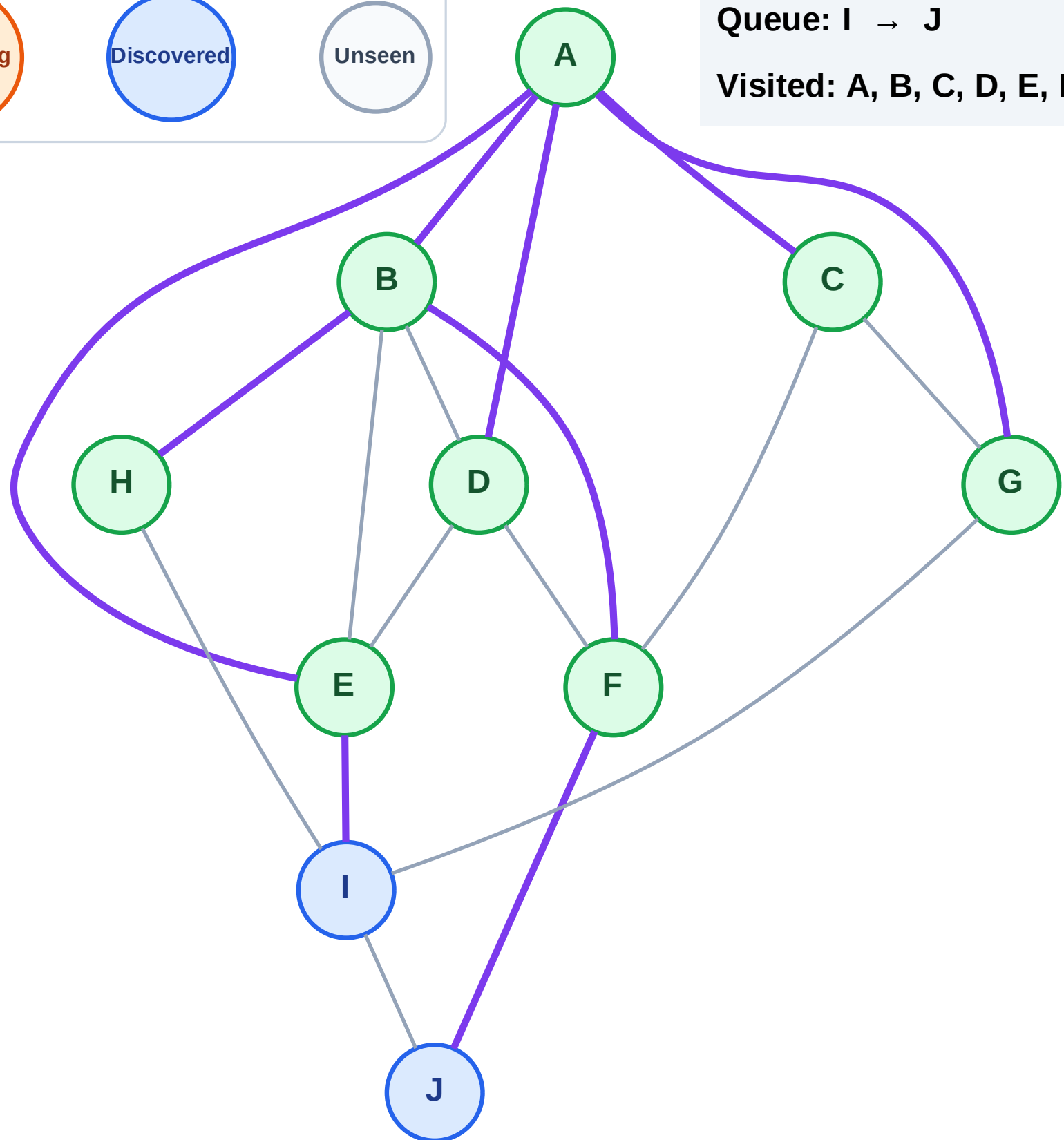
Step 17: No new neighbors from H

## Legend



Queue: I → J

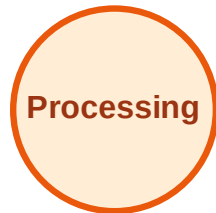
Visited: A, B, C, D, E, F, G, H



# BFS Traversal

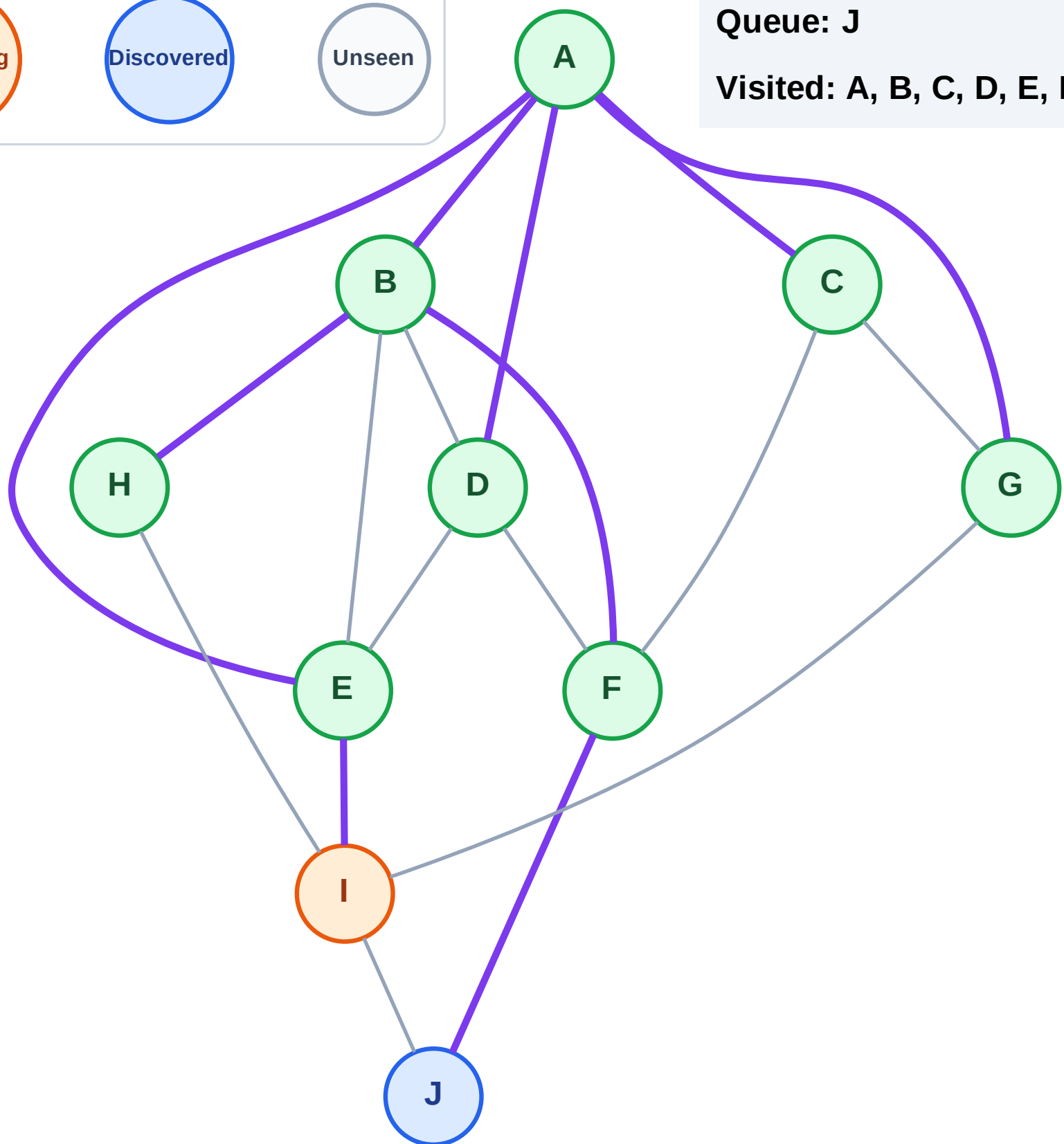
Step 18: Process node I

## Legend



Queue: J

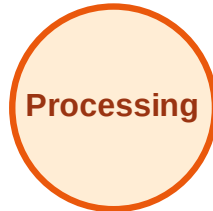
Visited: A, B, C, D, E, F, G, H



# BFS Traversal

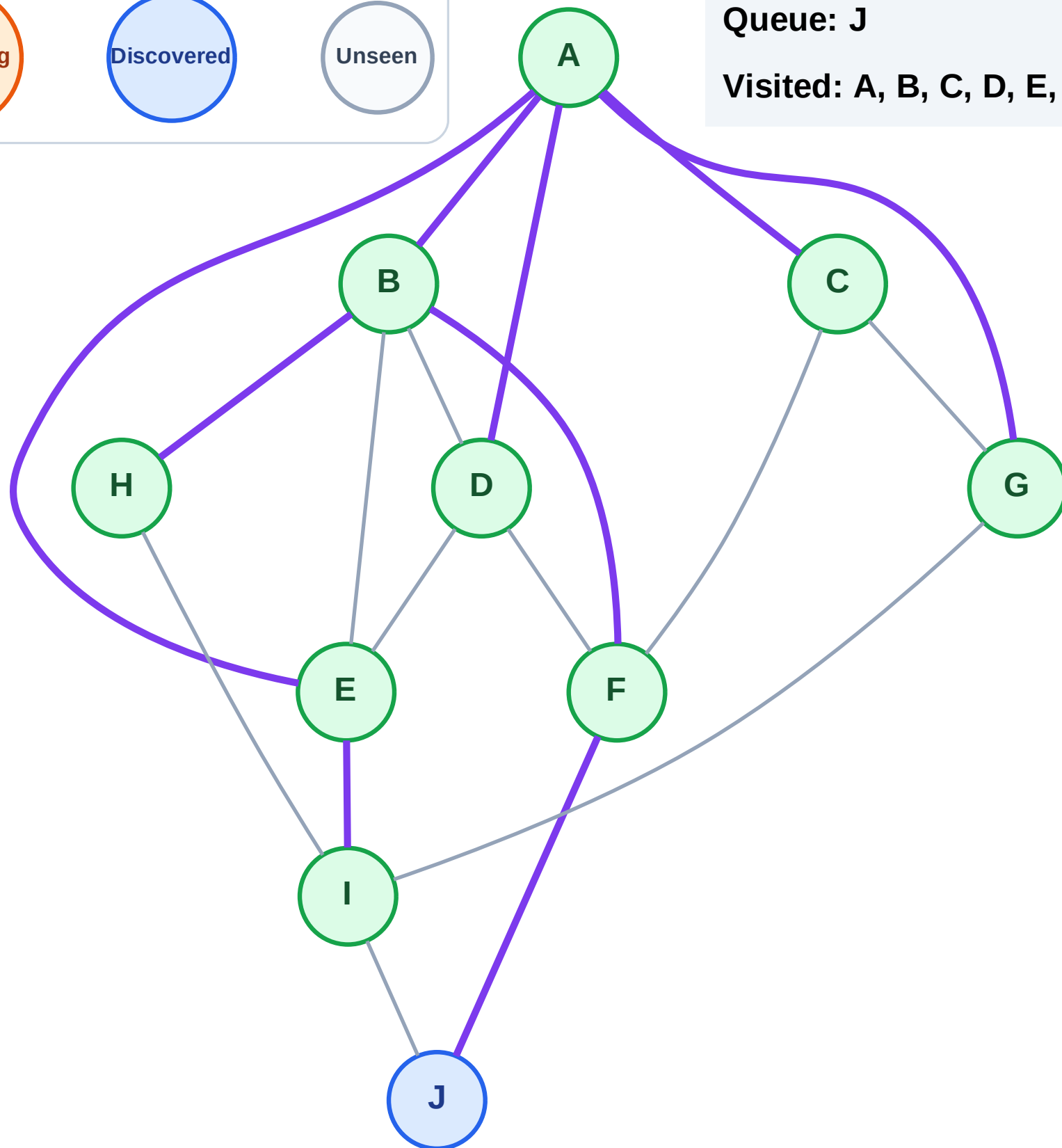
Step 19: No new neighbors from I

## Legend



Queue: J

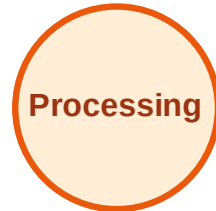
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

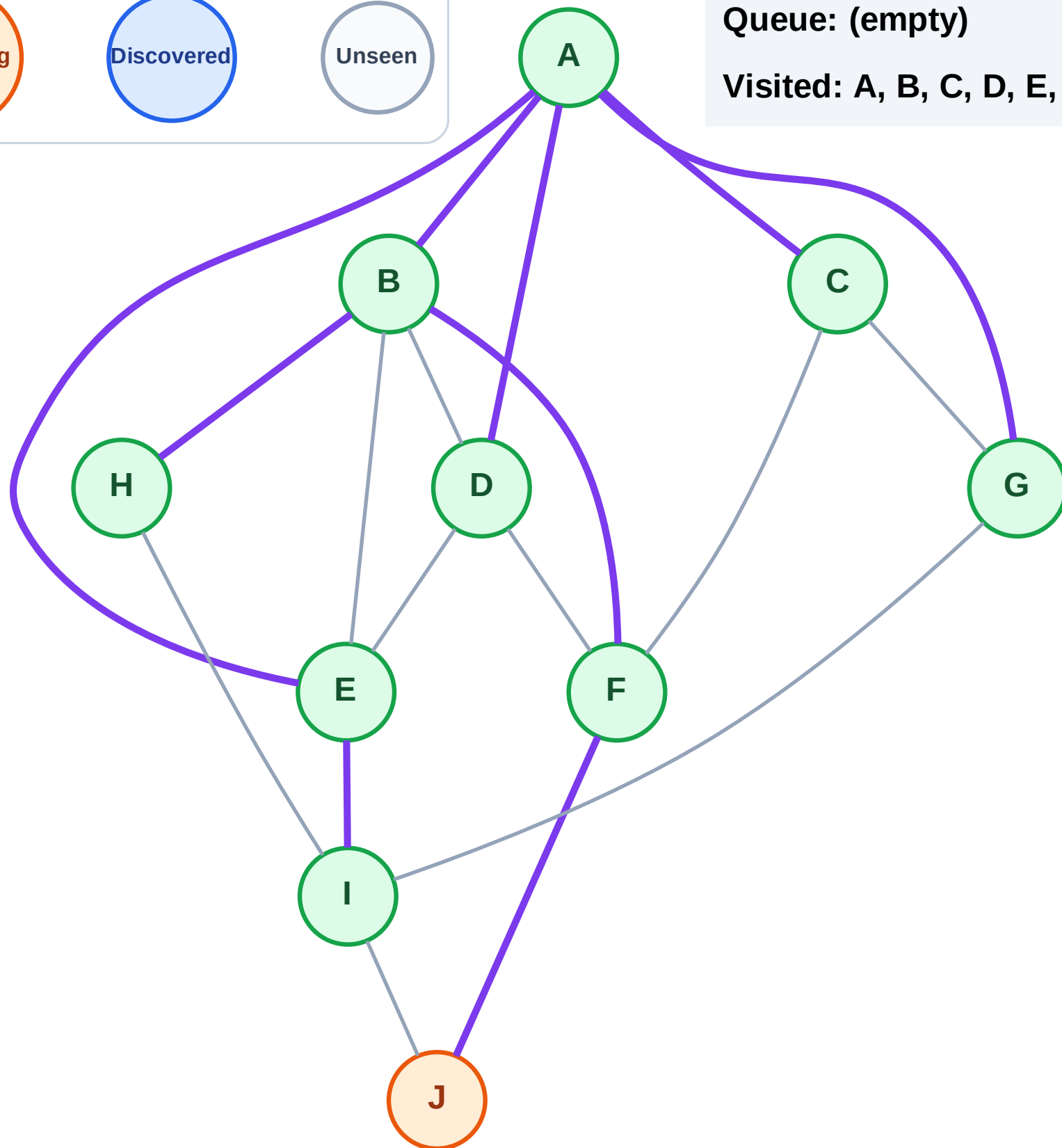
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

Step 21: No new neighbors from J

Legend

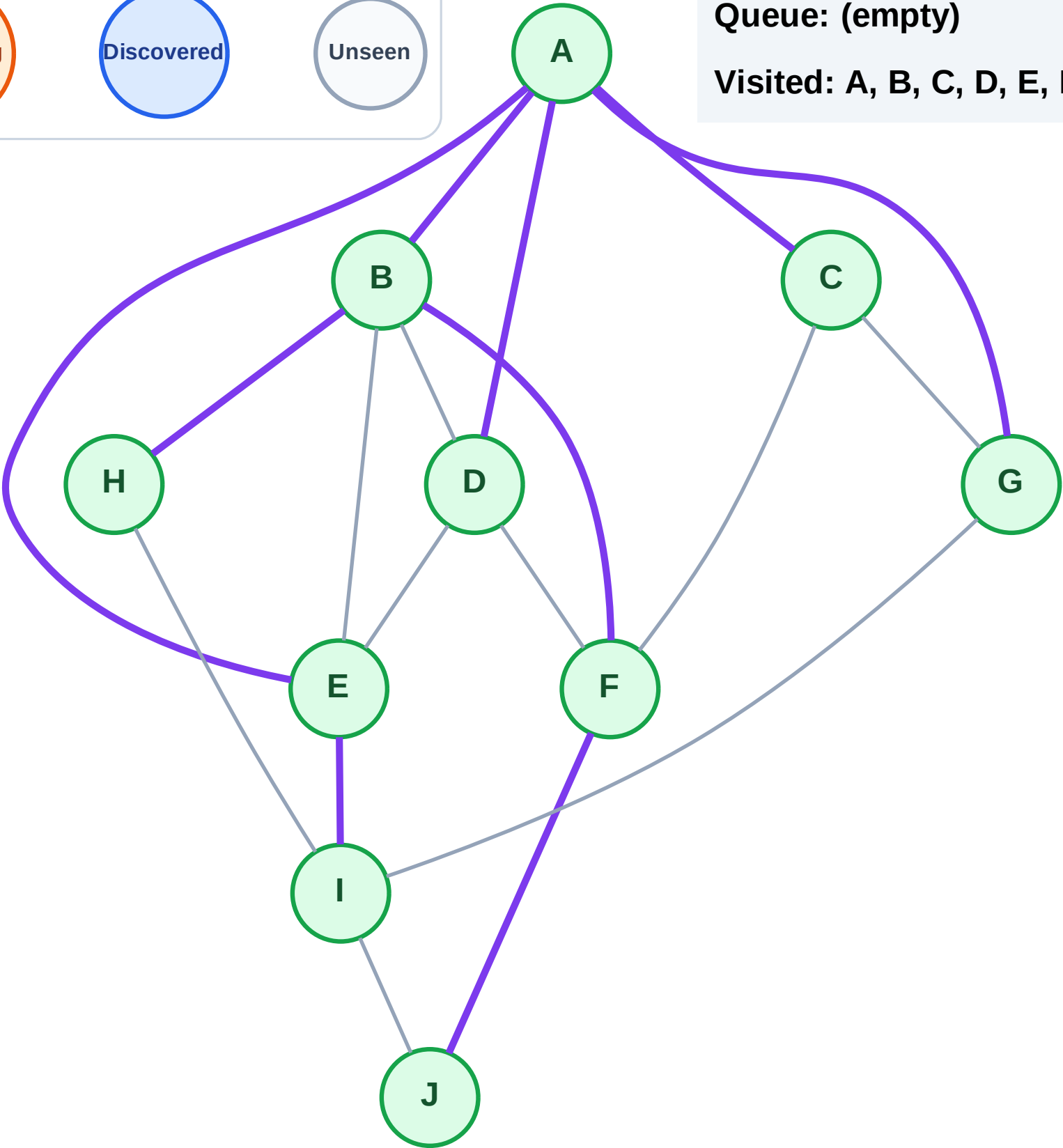
Complete

Processing

Discovered

Unseen

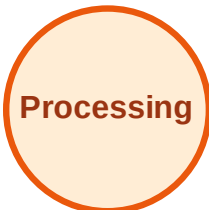
Queue: (empty)  
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

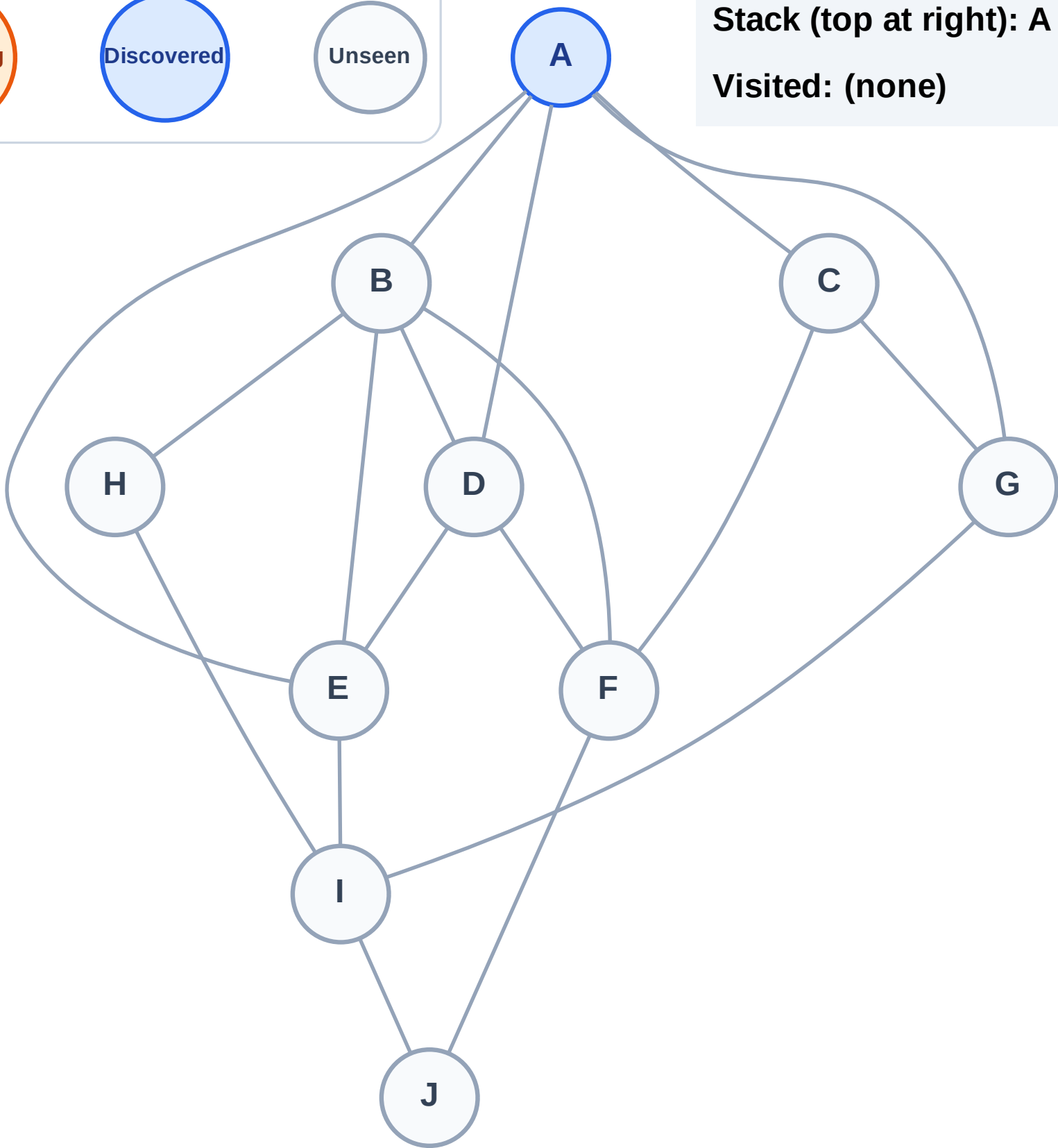
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

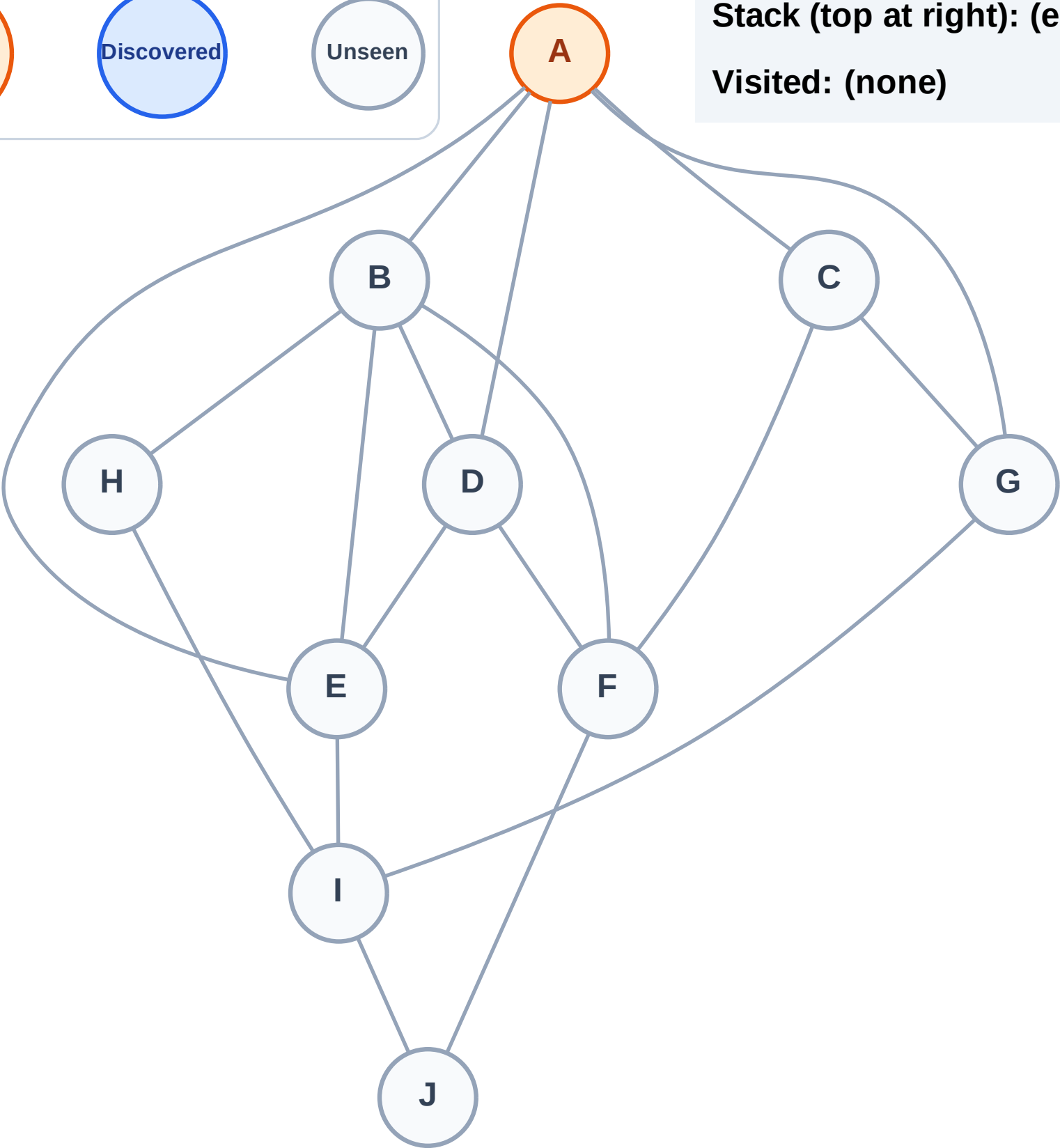
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)

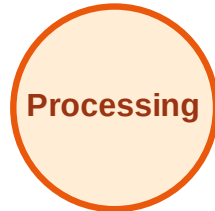




# DFS Traversal

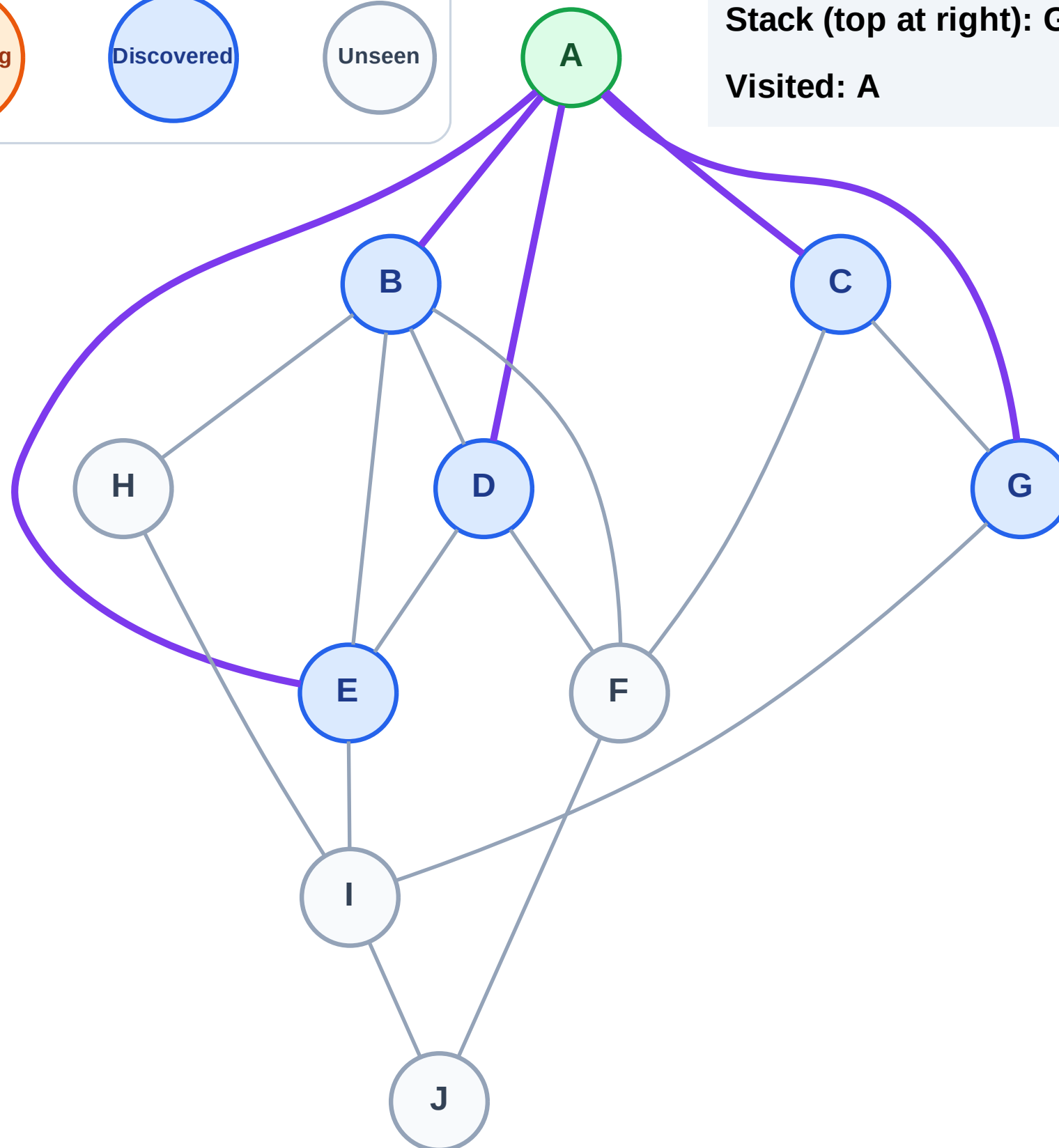
Step 3: Discover G, E, D, C, B from A

## Legend



Stack (top at right): G | E | D | C | B

Visited: A





# DFS Traversal

Step 4: Process node B

Legend

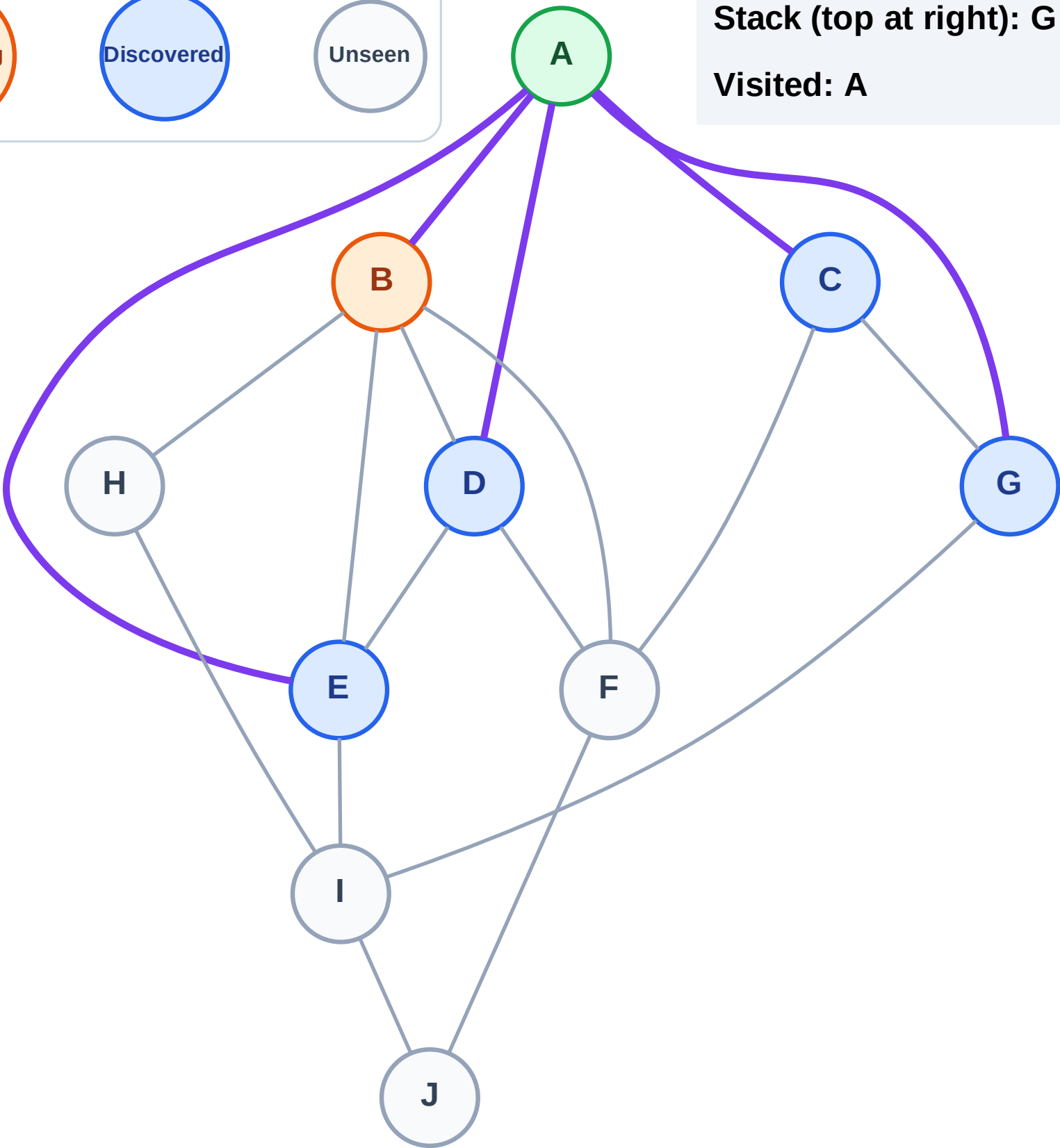
Complete

Processing

Discovered

Unseen

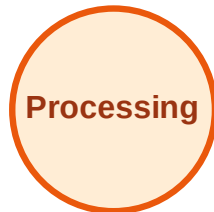
Stack (top at right): G | E | D | C  
Visited: A



# DFS Traversal

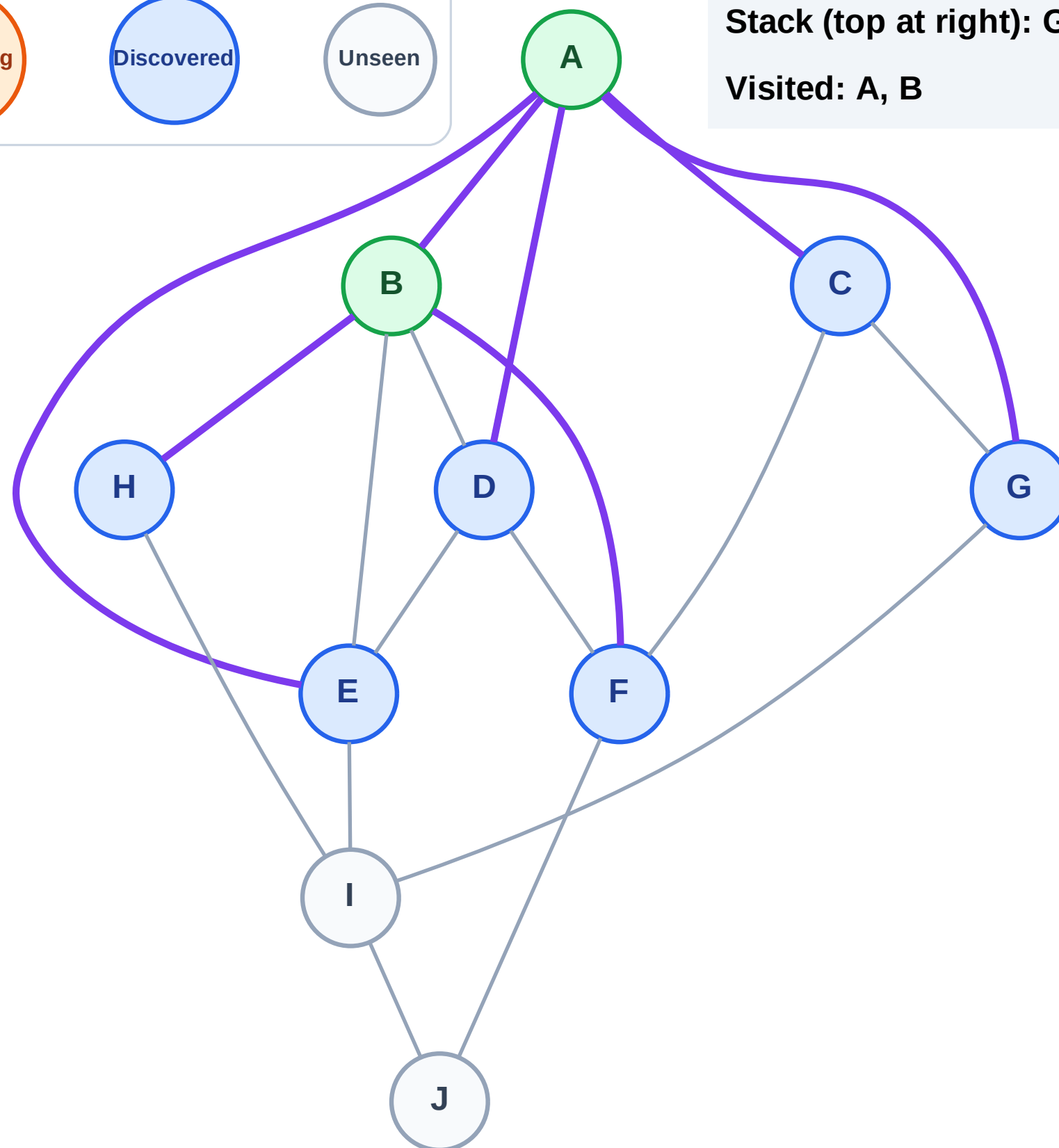
Step 5: Discover H, F from B

## Legend



Stack (top at right): G | E | D | C | H | F

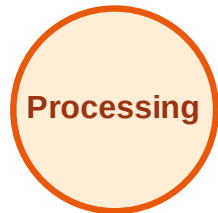
Visited: A, B



# DFS Traversal

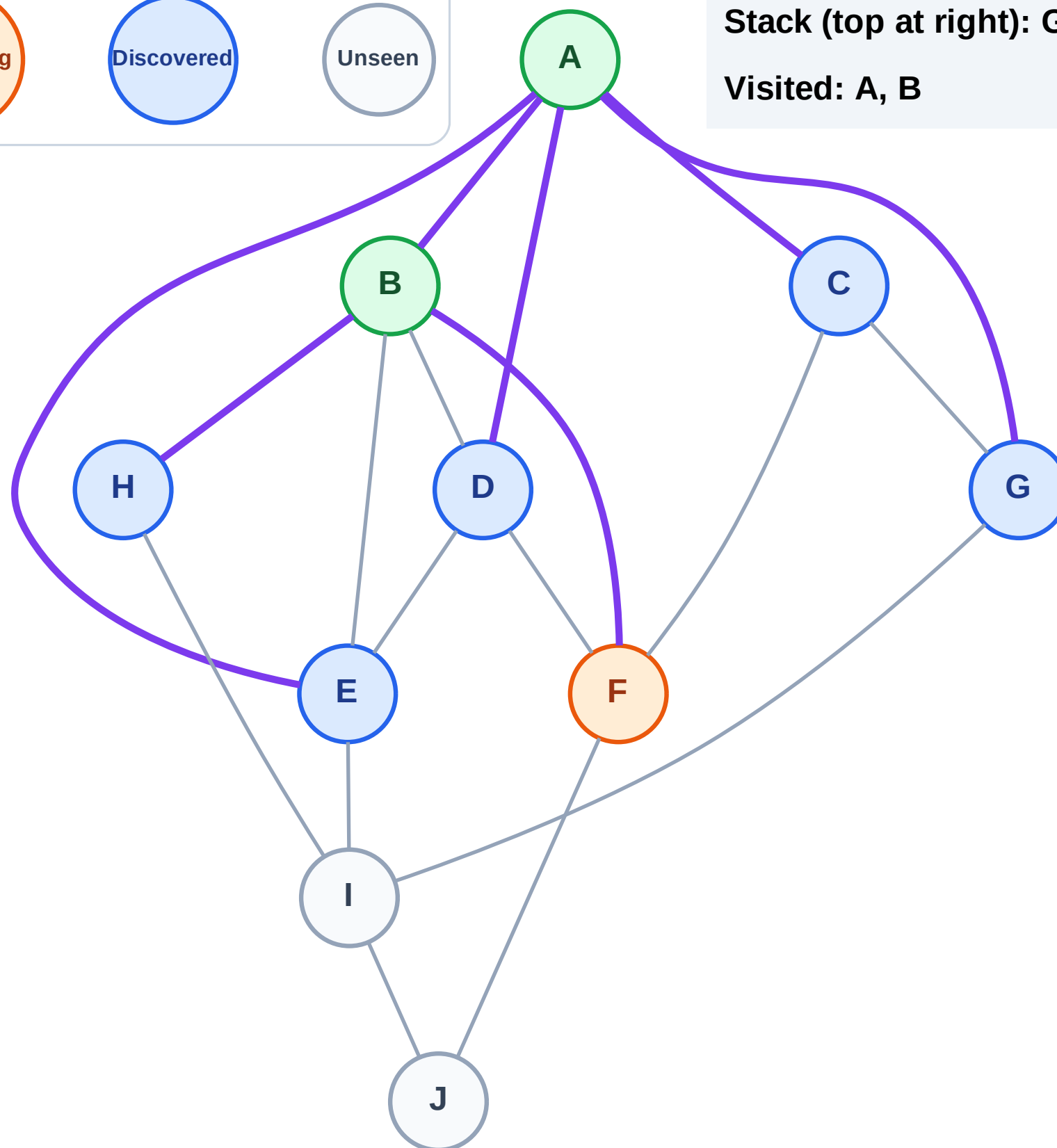
Step 6: Process node F

## Legend



Stack (top at right): G | E | D | C | H

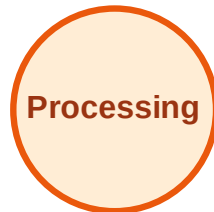
Visited: A, B



# DFS Traversal

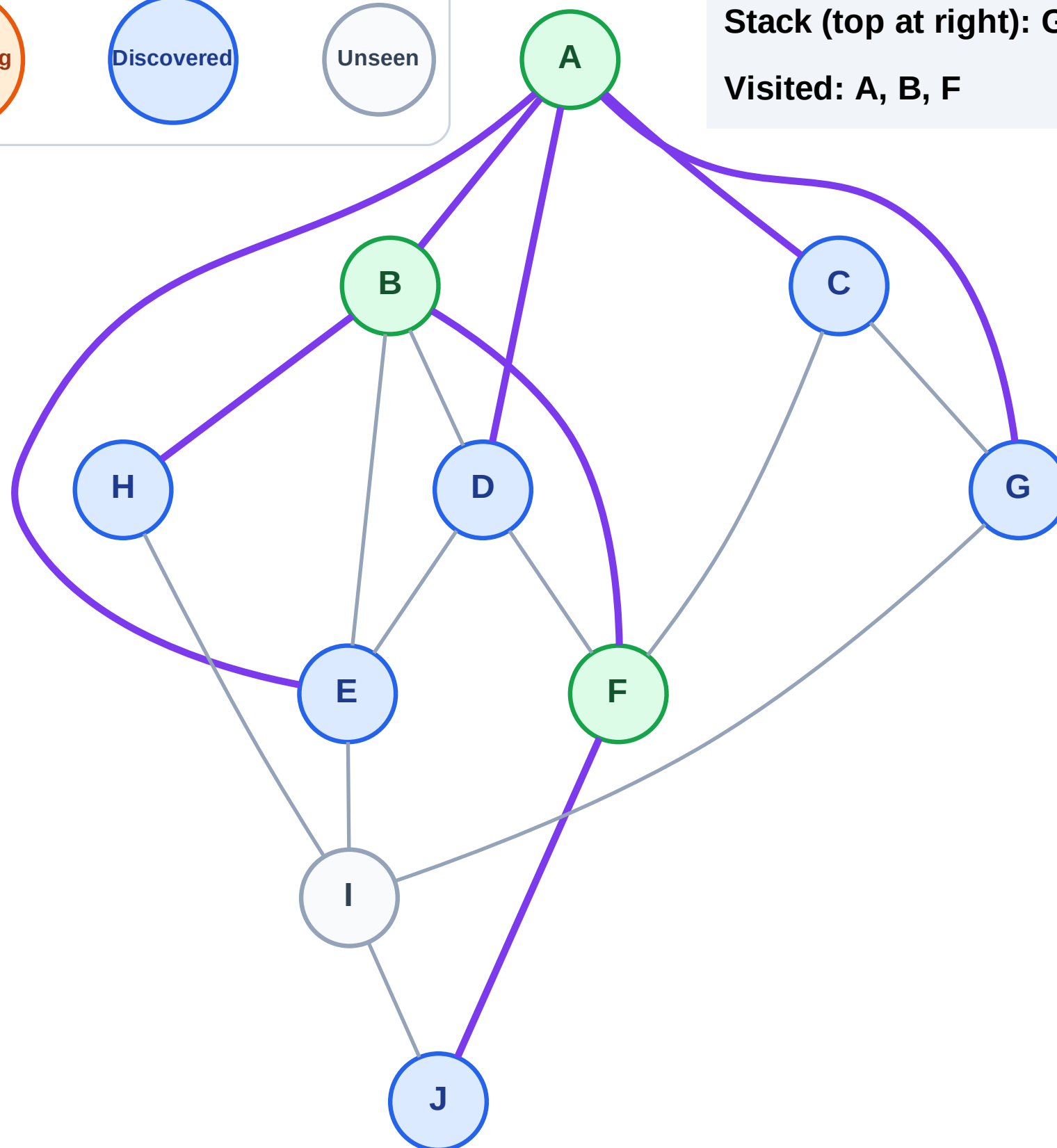
Step 7: Discover J from F

## Legend



Stack (top at right): G | E | D | C | H | J

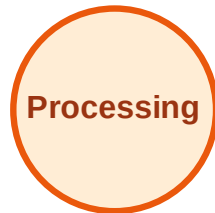
Visited: A, B, F



# DFS Traversal

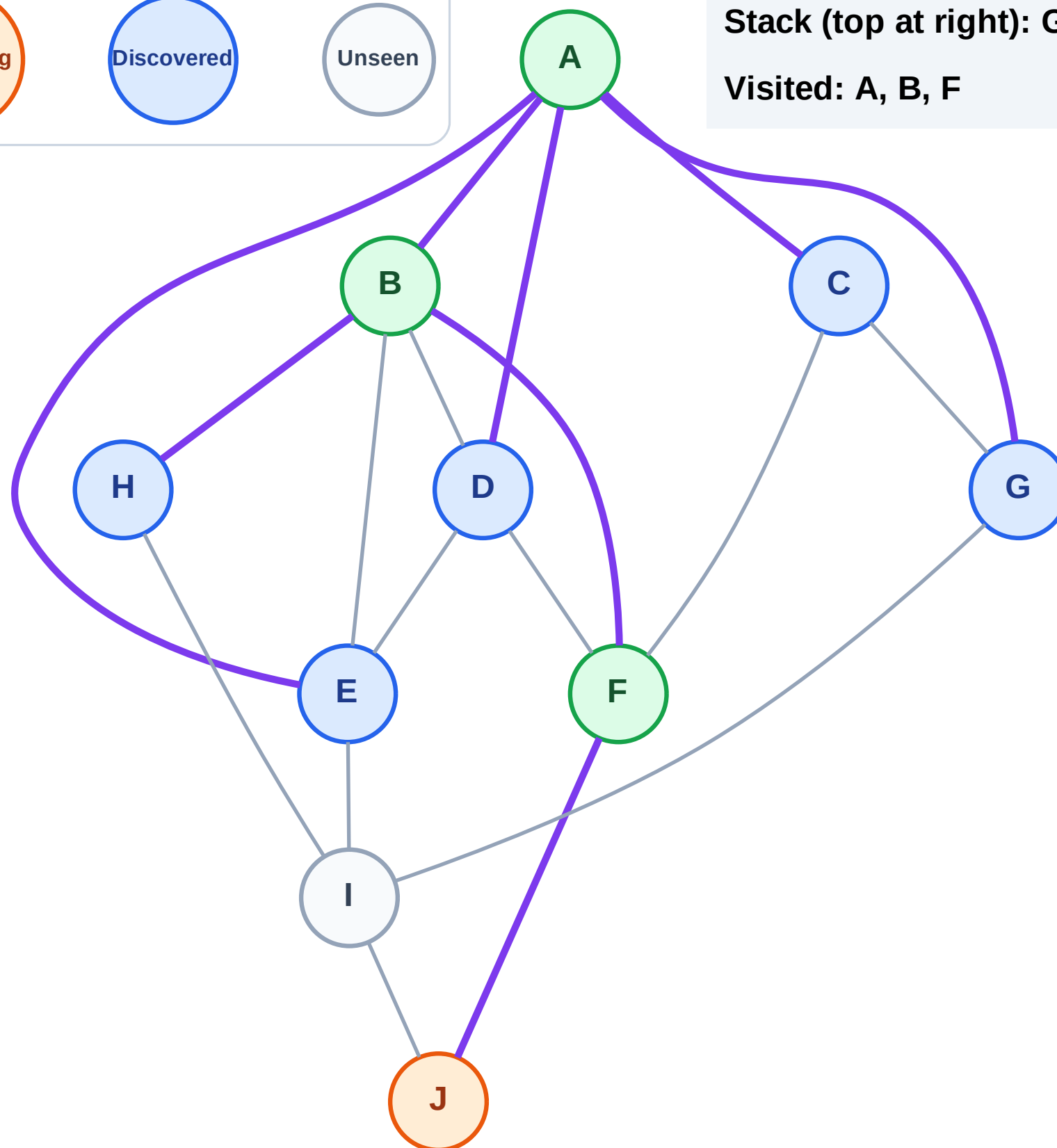
Step 8: Process node J

## Legend



Stack (top at right): G | E | D | C | H

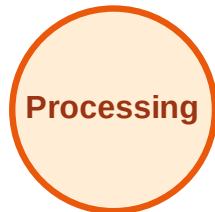
Visited: A, B, F



# DFS Traversal

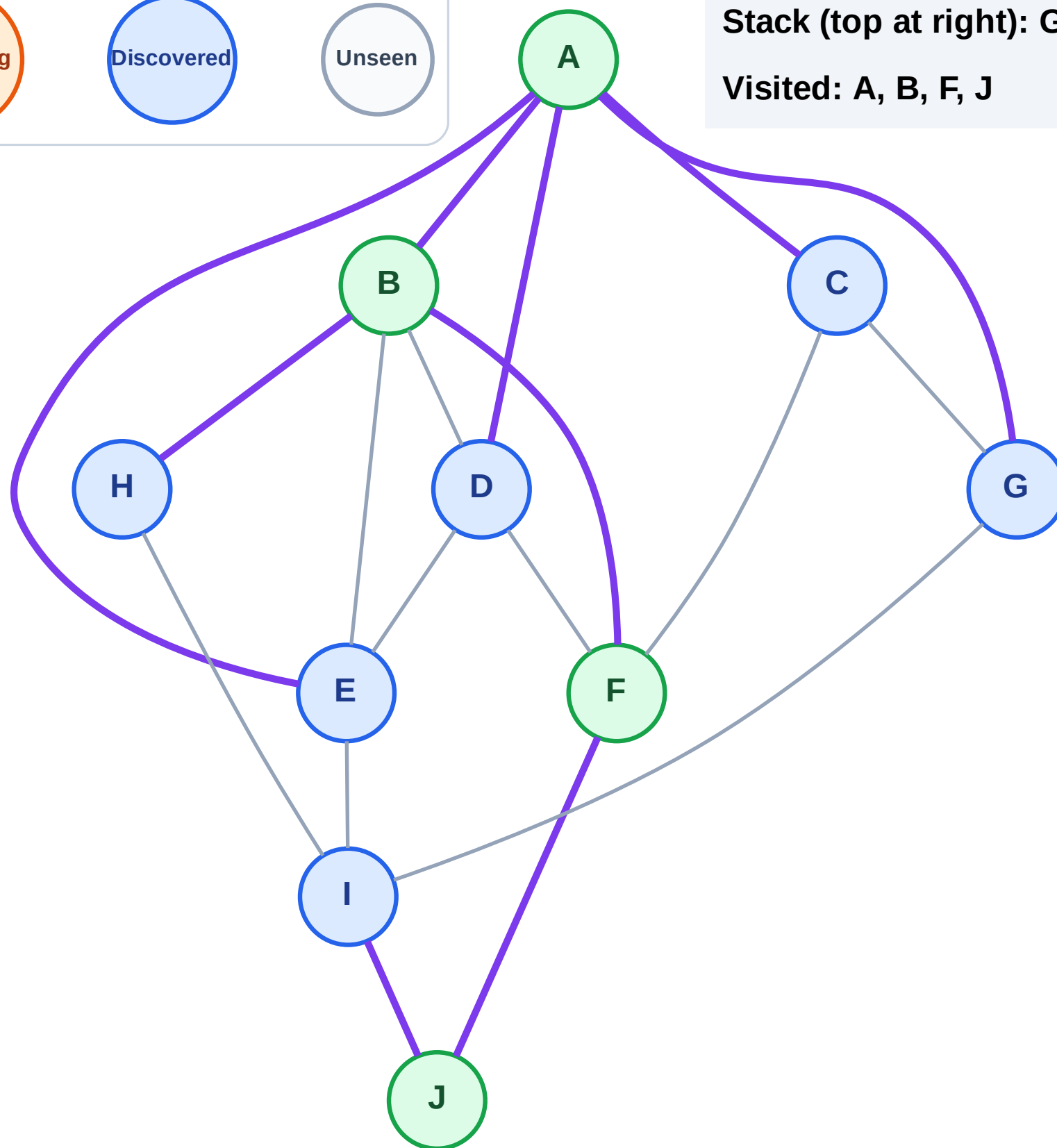
Step 9: Discover I from J

## Legend



Stack (top at right): G | E | D | C | H | I

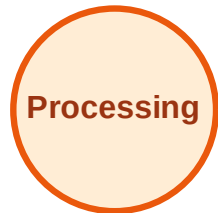
Visited: A, B, F, J



# DFS Traversal

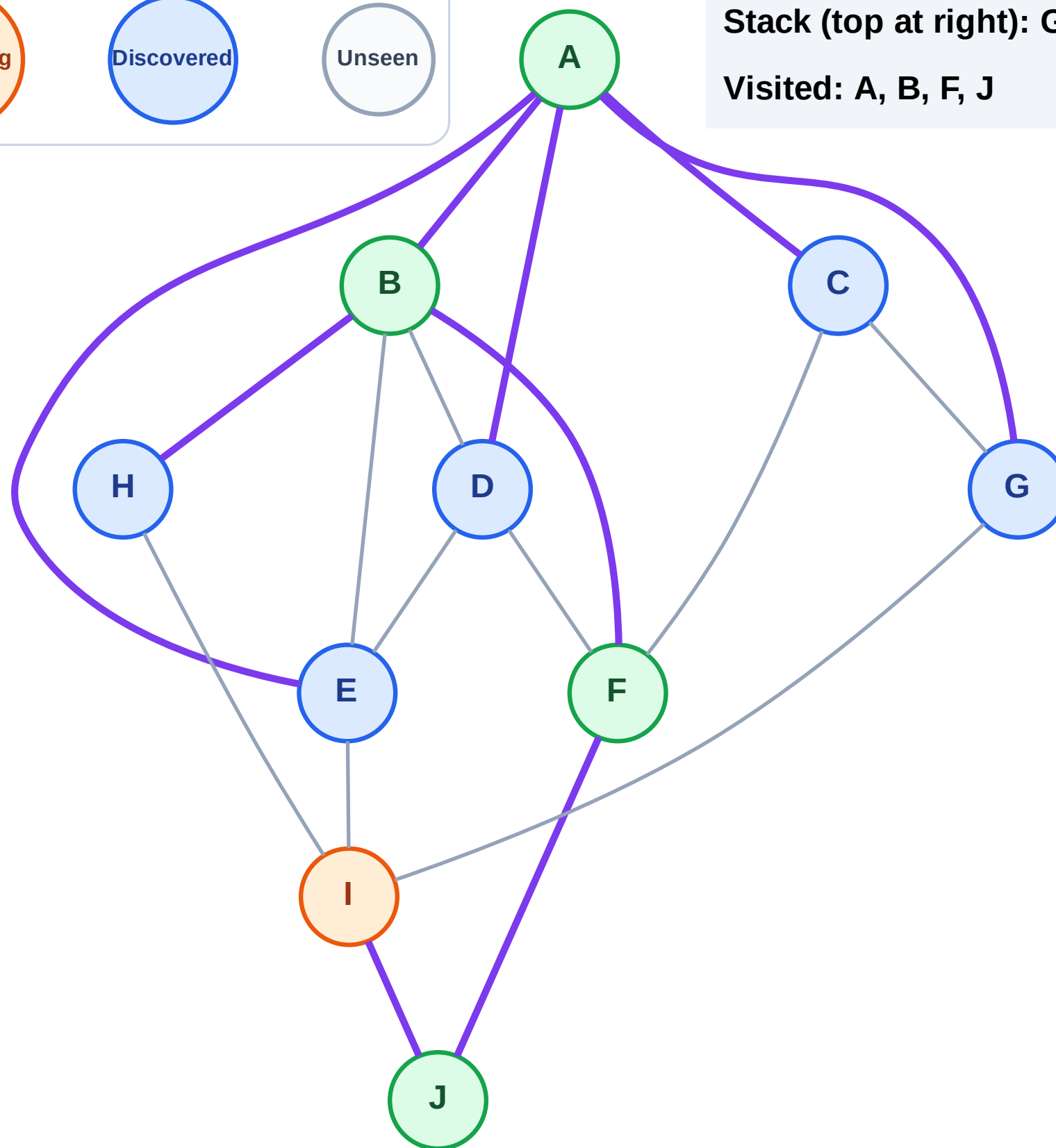
Step 10: Process node I

## Legend



Stack (top at right): G | E | D | C | H

Visited: A, B, F, J

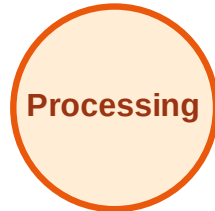




# DFS Traversal

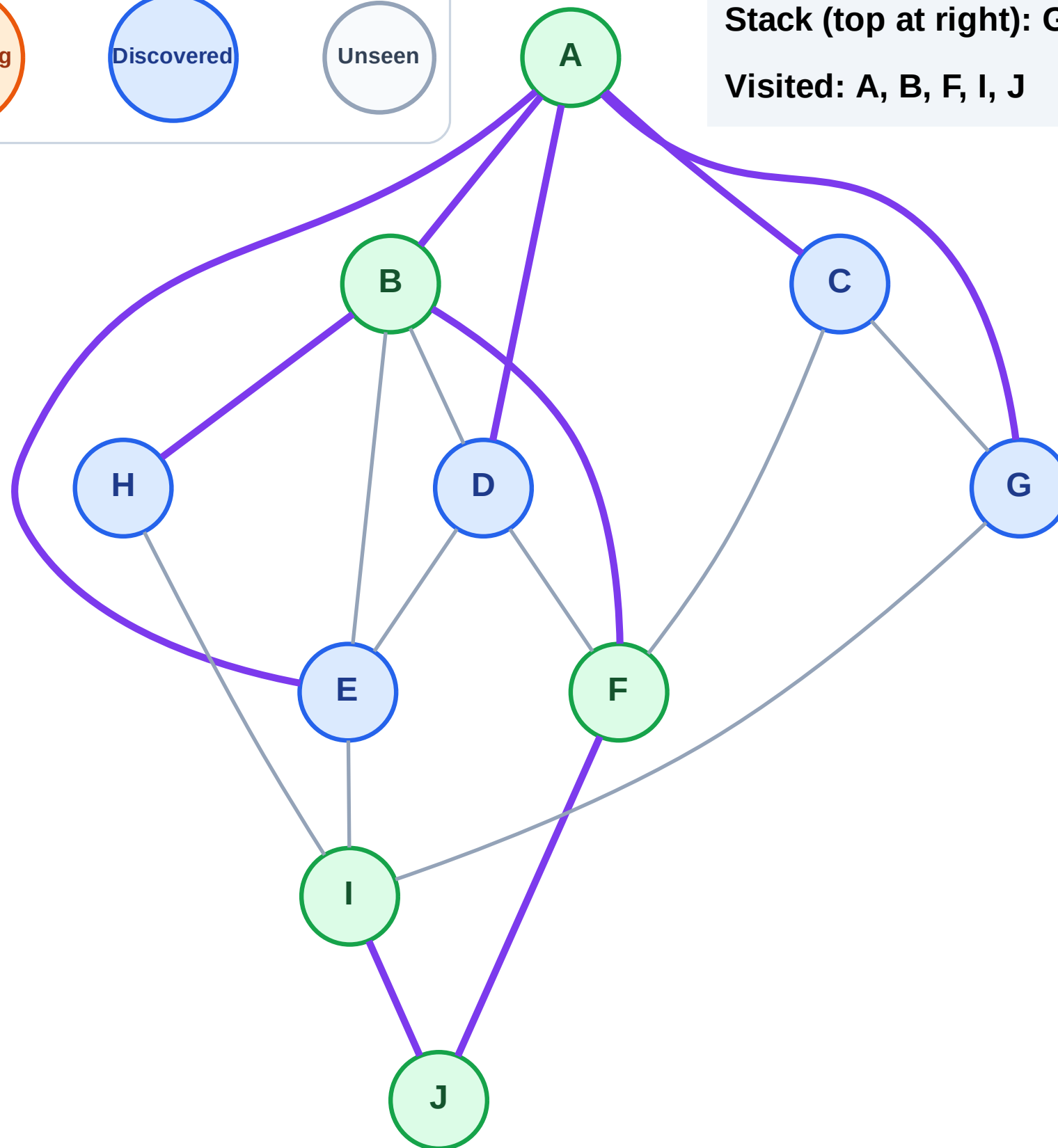
Step 11: No new neighbors from I

## Legend



Stack (top at right): G | E | D | C | H

Visited: A, B, F, I, J





# DFS Traversal

Step 12: Process node H

Legend

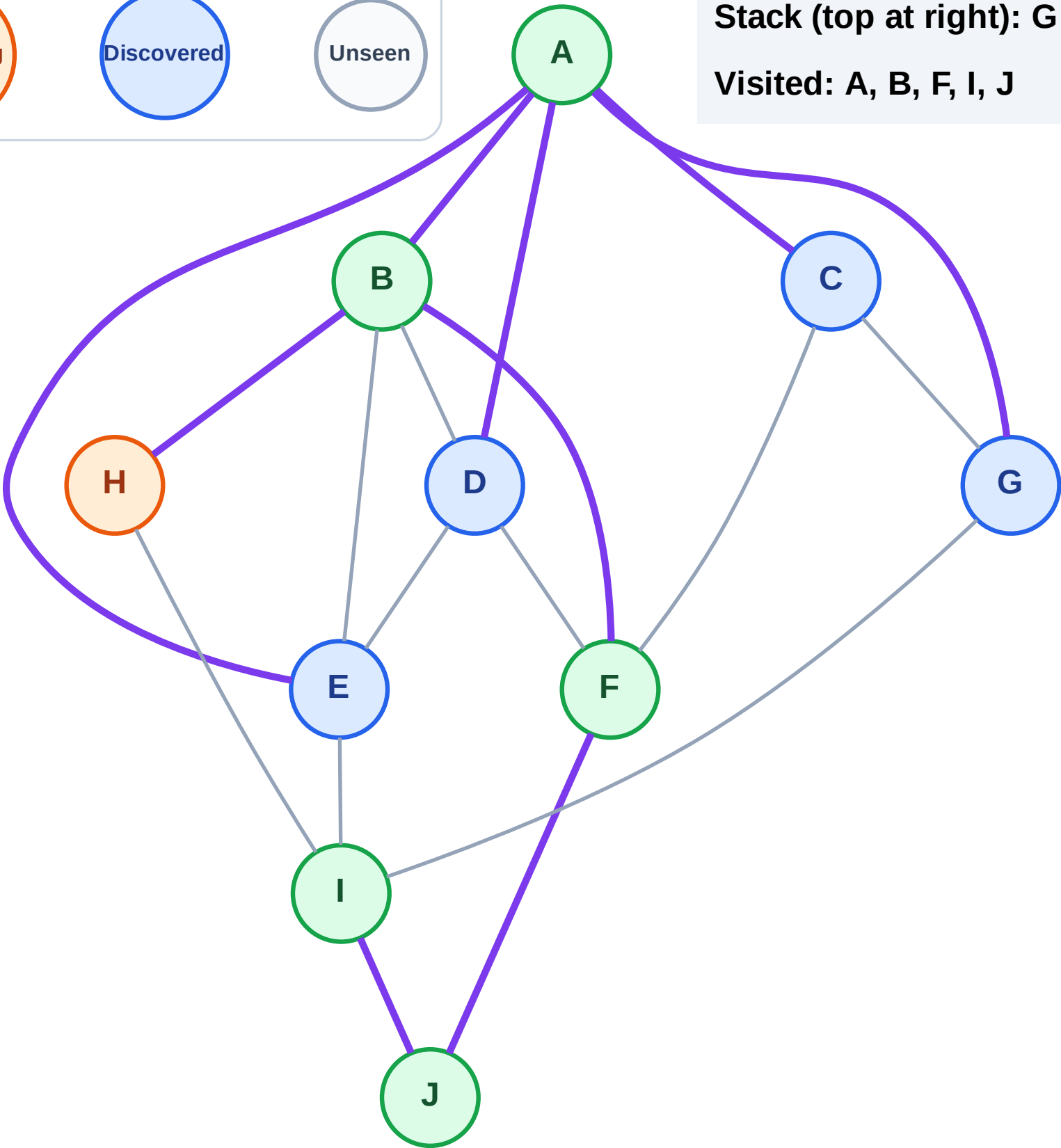
Complete

Processing

Discovered

Unseen

Stack (top at right): G | E | D | C  
Visited: A, B, F, I, J



# DFS Traversal

Step 13: No new neighbors from H

Legend

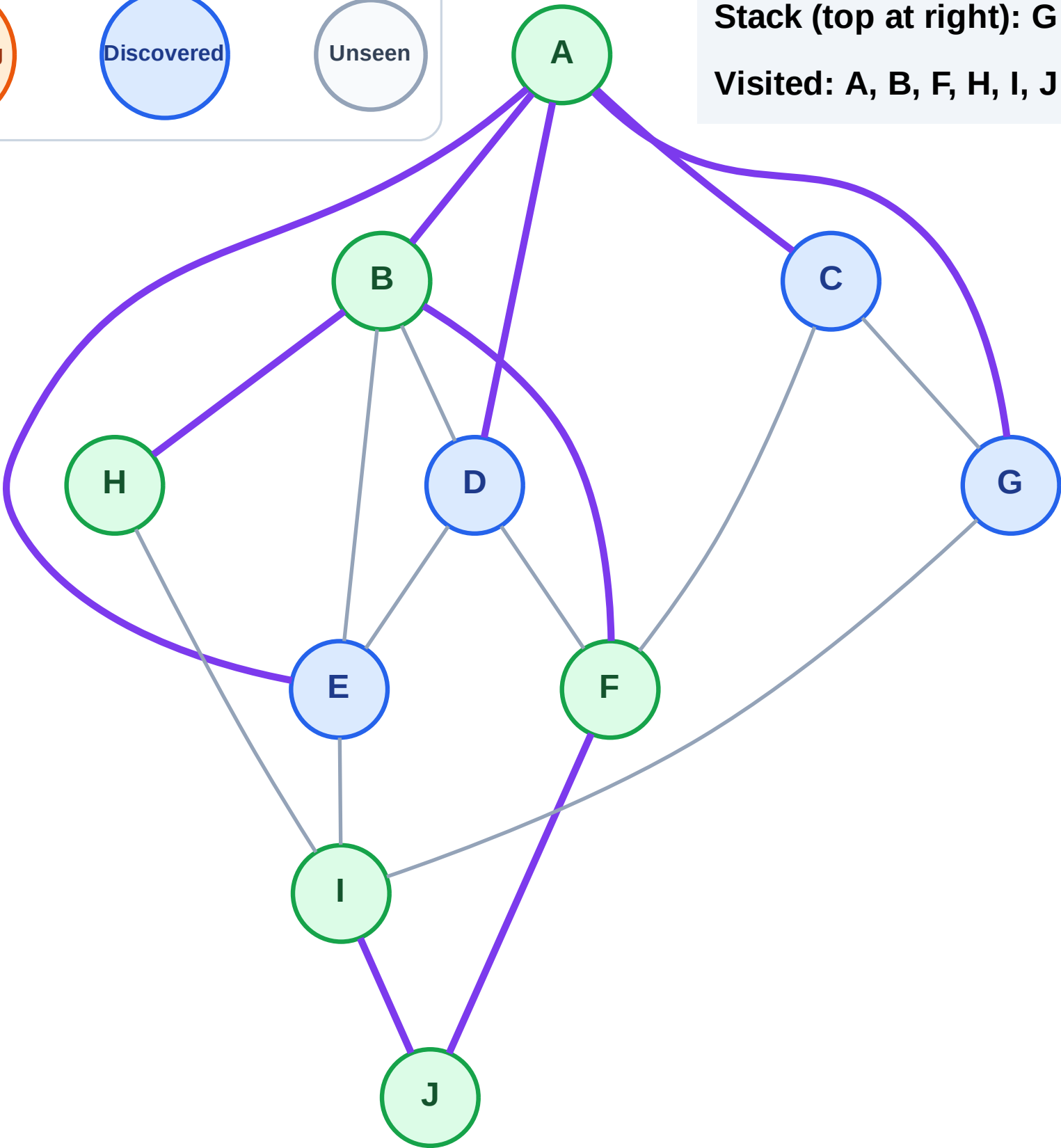
Complete

Processing

Discovered

Unseen

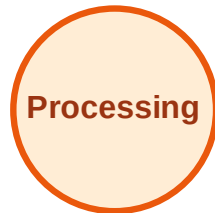
Stack (top at right): G | E | D | C  
Visited: A, B, F, H, I, J



# DFS Traversal

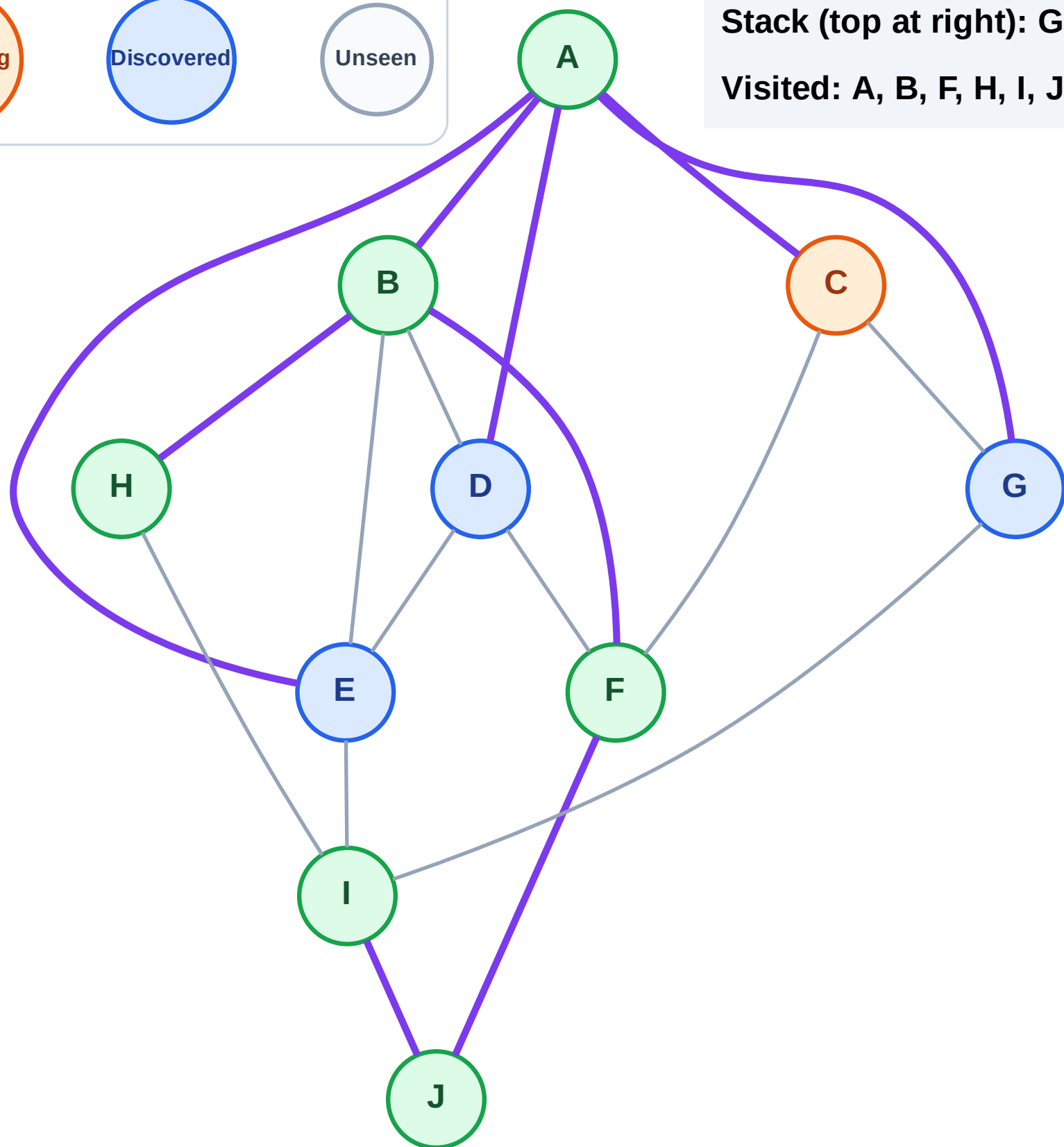
Step 14: Process node C

## Legend



Stack (top at right): G | E | D

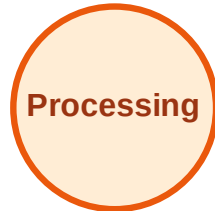
Visited: A, B, F, H, I, J



# DFS Traversal

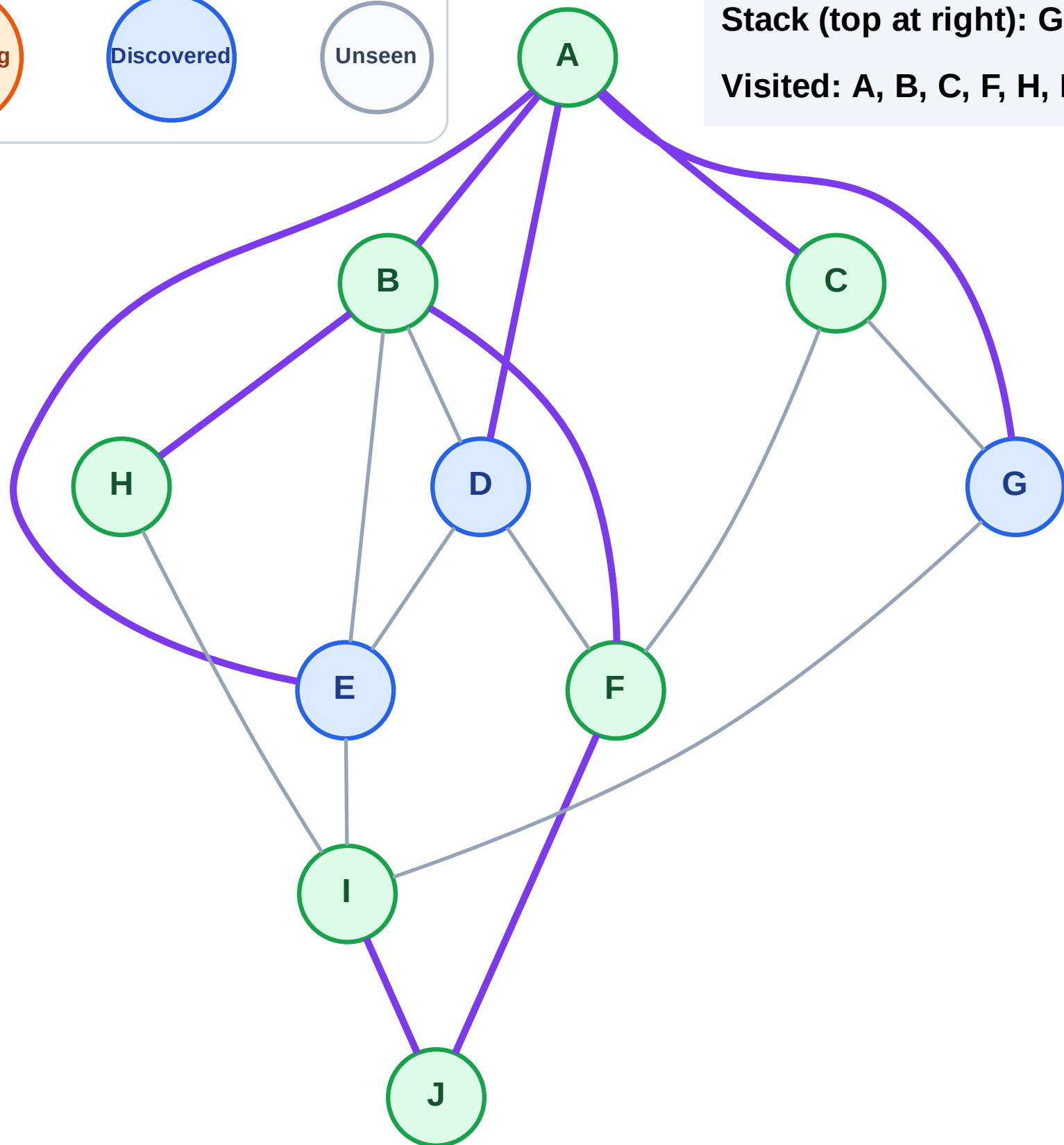
Step 15: No new neighbors from C

## Legend



Stack (top at right): G | E | D

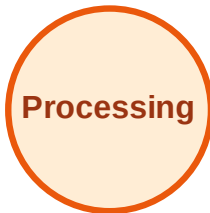
Visited: A, B, C, F, H, I, J



# DFS Traversal

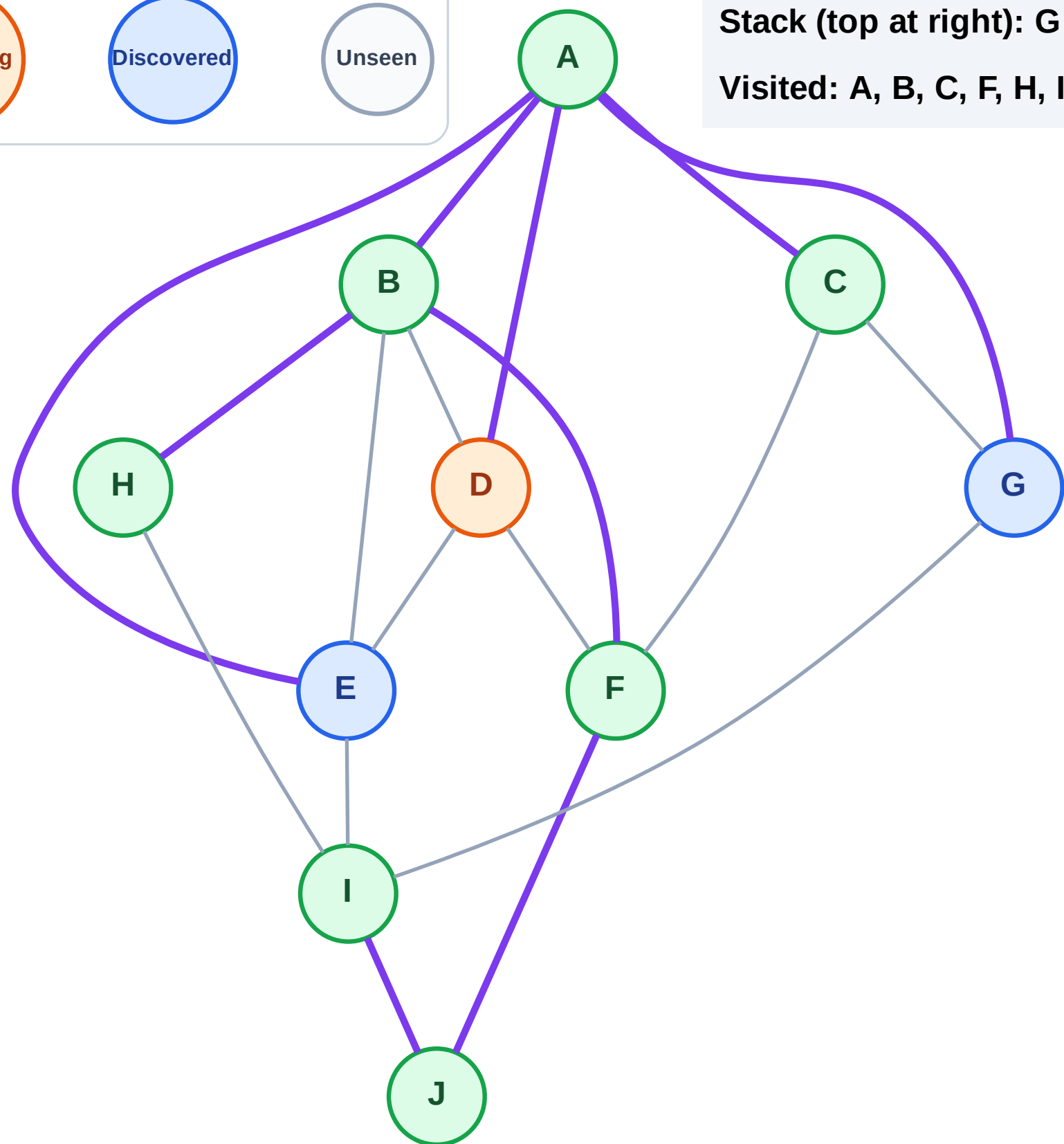
Step 16: Process node D

## Legend



Stack (top at right): G | E

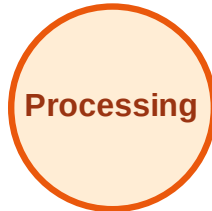
Visited: A, B, C, F, H, I, J



# DFS Traversal

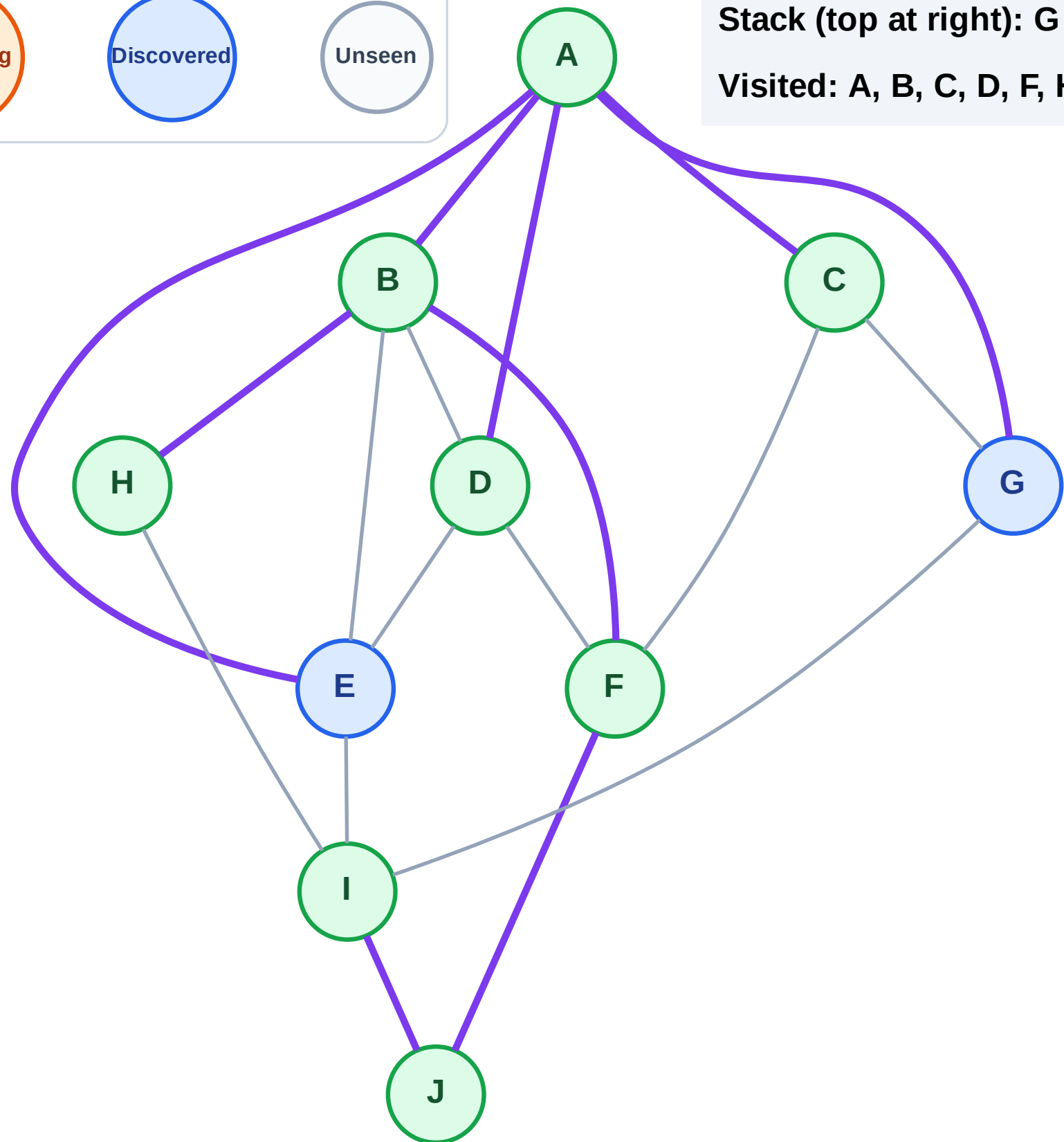
Step 17: No new neighbors from D

## Legend



Stack (top at right): G | E

Visited: A, B, C, D, F, H, I, J



# DFS Traversal

Step 18: Process node E

Legend

Complete

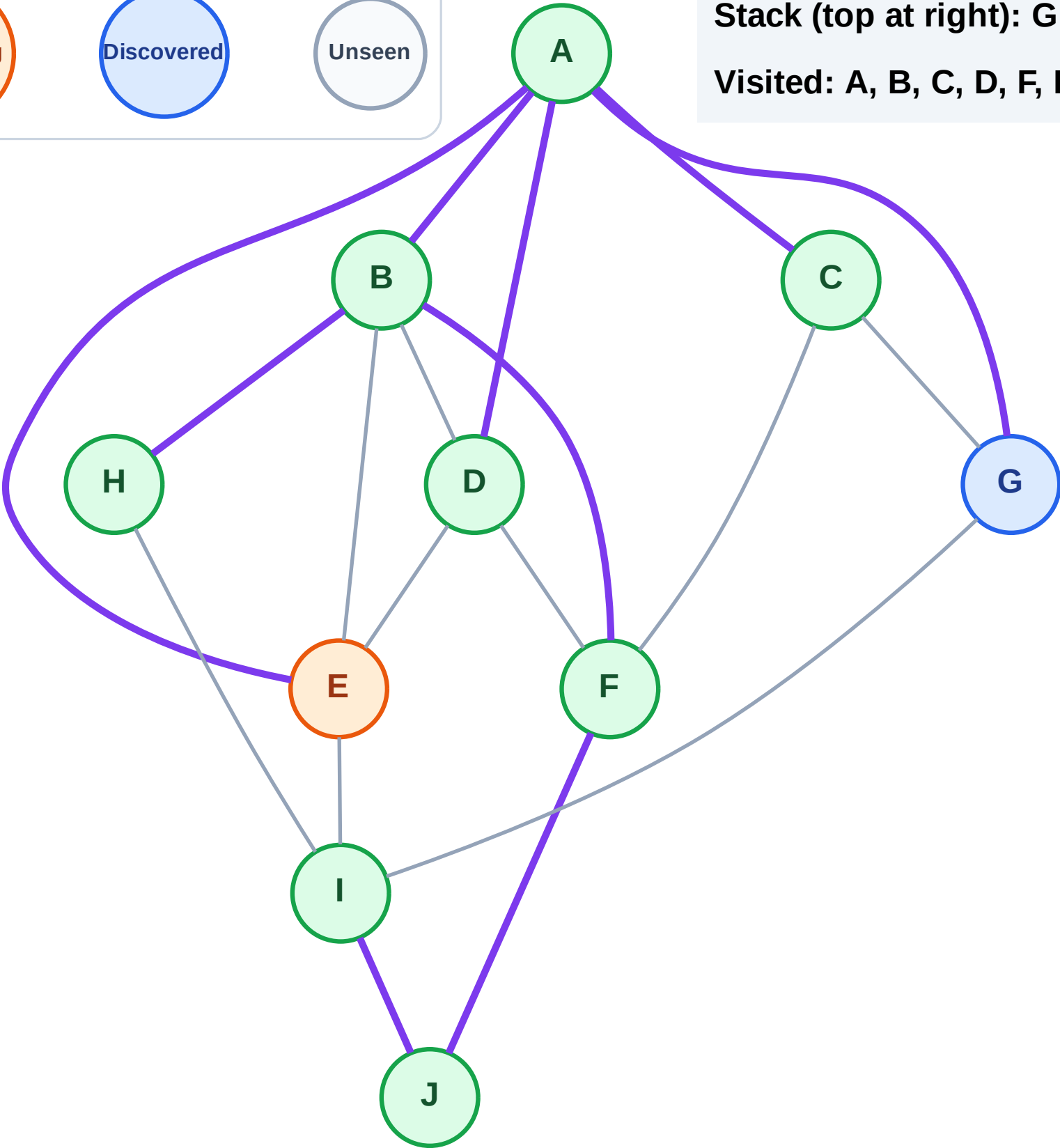
Processing

Discovered

Unseen

Stack (top at right): G

Visited: A, B, C, D, F, H, I, J

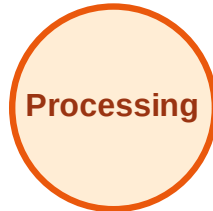




# DFS Traversal

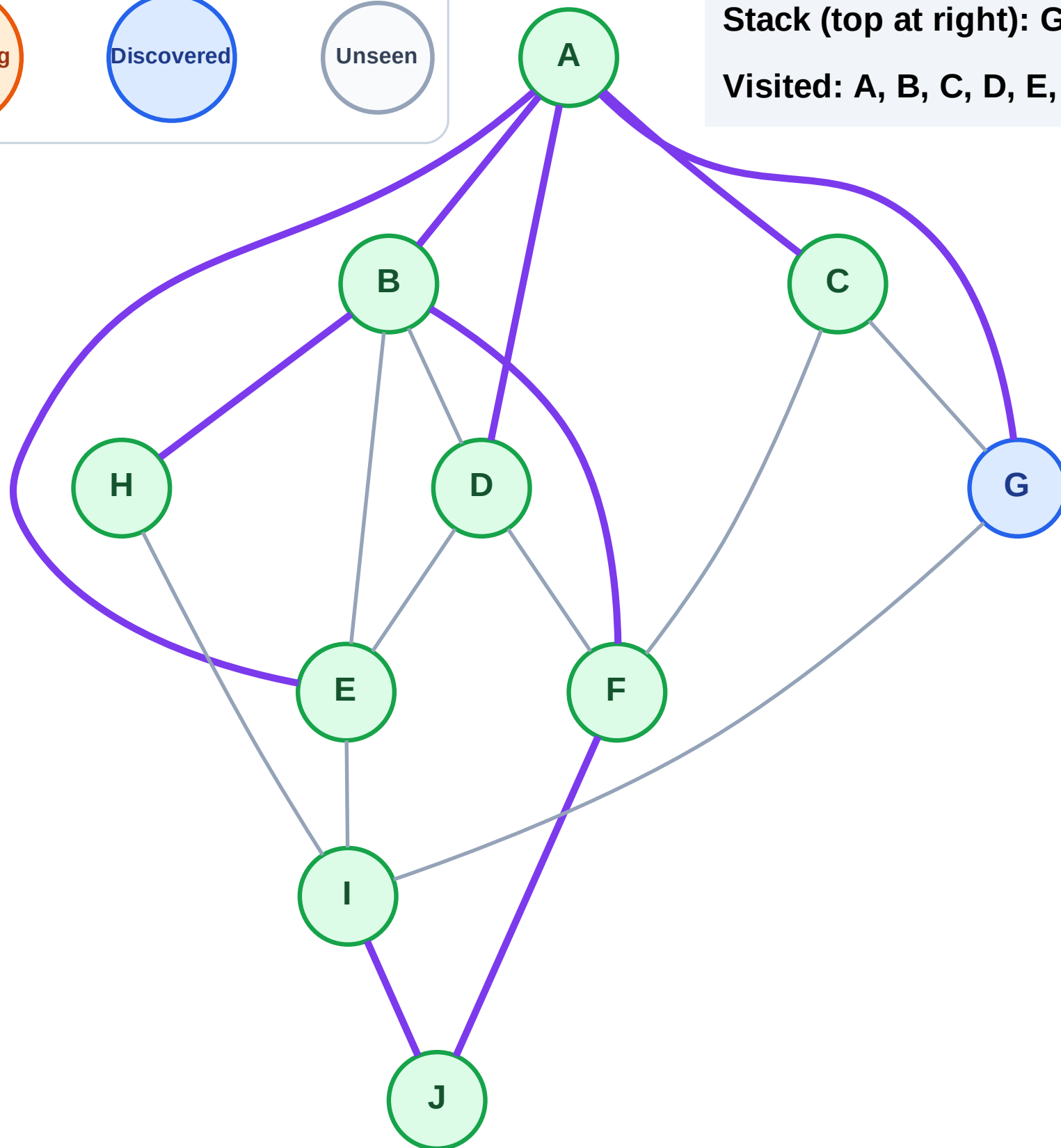
Step 19: No new neighbors from E

## Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J

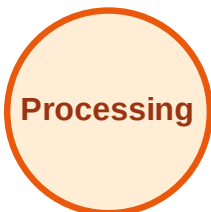




# DFS Traversal

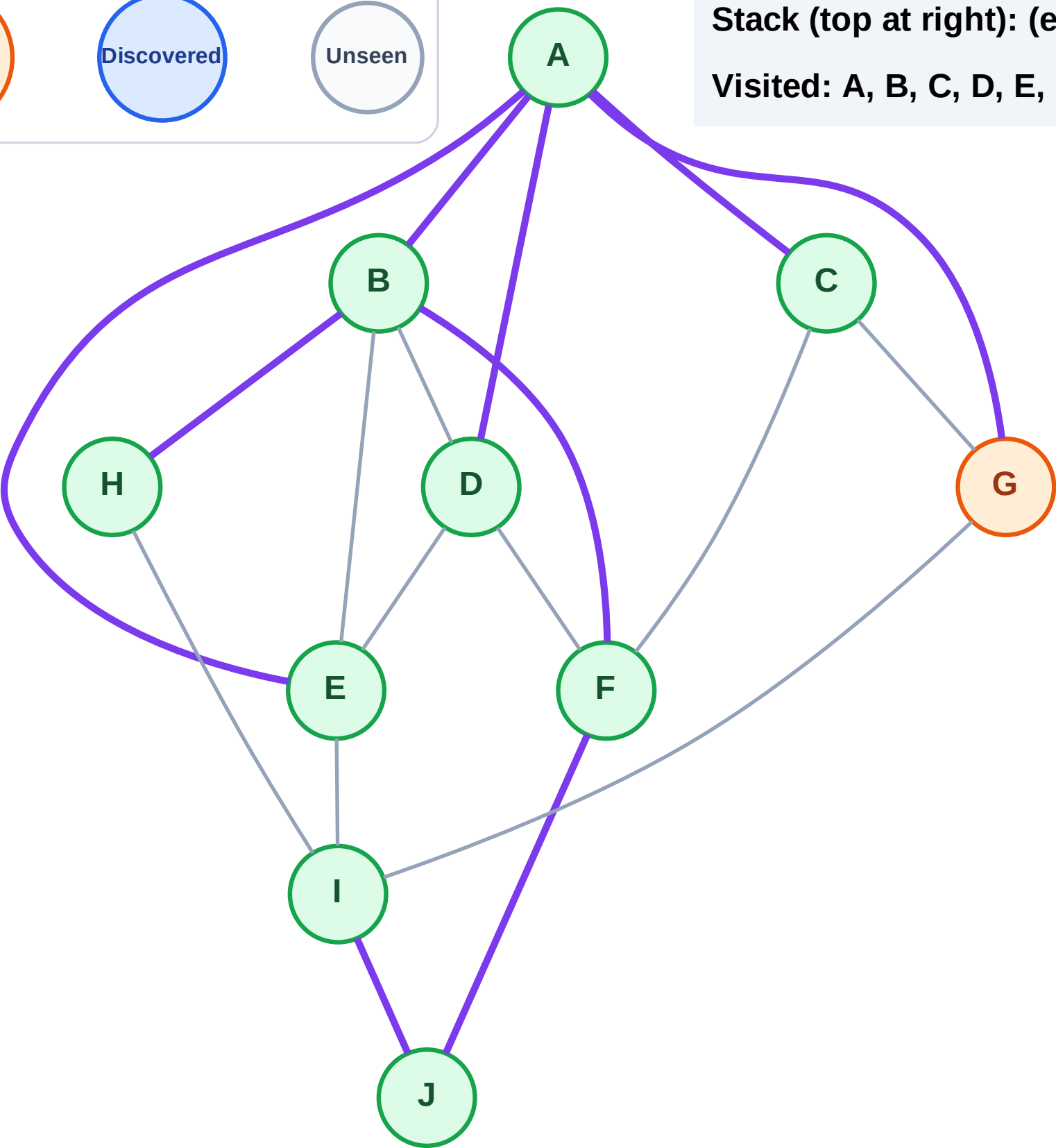
Step 20: Process node G

## Legend



Stack (top at right): (empty)

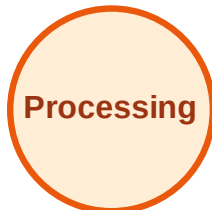
Visited: A, B, C, D, E, F, H, I, J



# DFS Traversal

Step 21: No new neighbors from G

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

